

Preferential trade agreements under policy uncertainty: theory and evidence from Brexit

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Abstract

When preferential trade agreements (PTAs) are renegotiated, tariff preferences become uncertain and trade could revert to the MFN regime. I develop a model of trade policy uncertainty within PTAs where firms choose the MFN or preferential regime. With uncertainty, more firms trade under MFN, with the option to access tariff preferences in the future. Trade values are reduced as more firms face the MFN instead of the preferential tariffs. Applying the model to Brexit, I find that ignoring the partial uptake of PTAs can bias estimates of uncertainty effects. Brexit uncertainty reduced UK's imports from PTA partners by 2.6%.

Keywords: Trade policy uncertainty; Trade agreements; Brexit.

JEL codes: JEL F13; F14; F68.

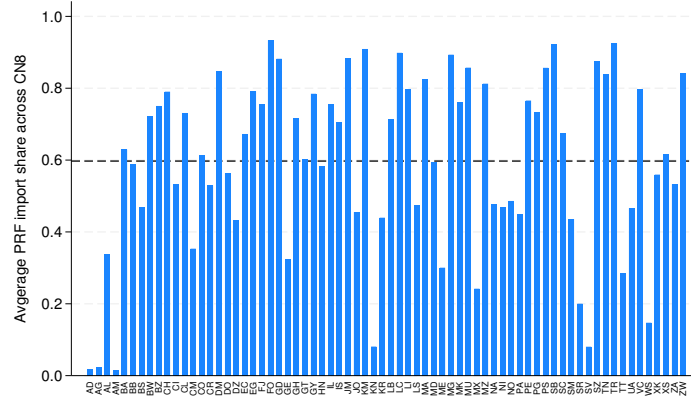
1 Introduction

According to UNCTAD (2023), in 2021 about half of world trade was done under some form of Preferential Trade Agreements (PTAs). PTAs have historically been a tool for securing market access and fostering international cooperation, and are generally believed to increase certainty for businesses as they signal commitments (Limão 2016; Carballo et al. 2022). However, in recent times trade policy uncertainty increased *within* PTAs, with episodes such as Brexit and the NAFTA renegotiation.

While current models of international trade can deal with different types of uncertainty, they are not suited to study policy uncertainty within PTAs. In particular, they do not recognise that, although PTAs

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Figure 1: Share of UK imports under preferential regime, 2013-15



Source: author's calculation based on Eurostat imports data for the UK and ITC MacMap tariff data for 2013-15. The figure shows the average share of preferential trade across CN 8-digit products with a positive MFN tariff. The horizontal line shows the average across countries.

grant tariff preferences, not all firms make use of them. How should we assess exposure to trade policy risk in PTAs when not all firms use preferences?

In this paper, I develop a model of trade policy uncertainty within PTAs, dealing explicitly with the partial uptake of tariff preferences. I provide an application to Brexit and the UK's trade with its PTA partners and demonstrate empirically the relevance of my model.

In PTAs firms can trade under the preferential (PRF) regime, which gives them access to lower customs tariffs compared to the Most Favoured Nation (MFN) rates, the default for members of the World Trade Organization. Despite tariff preferences being available, not all firms use them. Figure 1 shows the average preference utilisation rate across products for UK imports from PTA partners. The average preference utilisation rate is 58% with great variation across countries, even among major exporters. For instance, the average rate is 93% for Turkey, 49% for Norway and as low as 25% for Mexico.¹ The partial uptake of tariff preferences suggests the presence of some additional costs related to the utilisation of PTAs. Such costs can be dealing with bureaucratic procedures or compliance with Rules of Origin, for instance, and they create trade regime heterogeneity.

What happens if a PTA is renegotiated and tariff preferences become uncertain? As exporting involves an initial sunk cost, uncertainty can deter firms from starting to export (Handley and Limão 2022). The prospect of losing tariff preferences, coupled with the fixed costs associated with their utilisation, can

¹The share is computed on Combined Nomenclature (CN) 8-digit products with non-zero MFN tariff over 2013-15, and excluding processing trade which is subject to tariff rebates.

dissuade firms from exporting.

However, the threat of losing preferential tariffs will not affect all exporters, but only those that utilise them. Applying a standard model of trade policy uncertainty and assuming that all firms utilise tariff preferences would result in overstating the effects of uncertainty in PTAs.

I introduce a model of trade between PTA partners under uncertainty, wherein accessing preferential tariffs entails an additional fixed cost compared to the standard MFN regime, in line with the literature on the utilisation of tariff preferences. The model features trade regime heterogeneity, with some firms trading under the MFN and others under the preferential regime. The model predicts that when preferential tariffs become uncertain, only firms that would choose the preferential regime in the absence of uncertainty are concerned with the potential loss of preferential tariffs. Under uncertainty, these firms choose to export under the MFN regime with the option to switch to the preferential one in the future. As more firms pay the MFN rather than the PRF tariff under uncertainty, export values are reduced.

Trade policy uncertainty is modelled as in Handley and Limão (2017) (henceforth HL), but applied to the model with heterogeneous tariff regimes. While in HL, which features one tariff regime only, uncertainty reduces entry into exporting, in my model uncertainty about preferential tariffs reduces entry into the preferential regime, but firms might still trade under the MFN regime.

I show how to derive a theory-consistent measure of risk in PTAs at the product level based on the model with regime heterogeneity. This risk measure is the share of imports under the preferential regime (exposure to the risk) multiplied by the margin between the preferential and MFN tariff (the potential profit loss). I derive an empirical equation that nests the one of the HL model under the assumption of a single trade regime. As in HL, the effect of uncertainty on exports is proportional to the PRF/MFN tariff margin, but we now have to account for the preference utilisation rate as well.

I test the model using an excellent natural experiment: the UK's trade with its PTA partners following the 2016 Brexit referendum on leaving the European Union (EU). By leaving the EU, the UK was also leaving trade agreements with about 70 countries beyond the EU, generating a concrete possibility for the UK's trade with PTA countries to revert to the MFN regime. To counteract this uncertainty, the UK government started the process of rolling over existing PTAs by signing Continuity Agreements (CAs) in 2019, two and a half years after the Brexit referendum.

Empirical results show that the uncertainty generated by the 2016 referendum negatively affected the UK's trade with its PTA partners. Continuity Agreements mitigated the negative effect of uncertainty but did not offset it completely, reducing uncertainty by a third on average. I find differences between the empirical results of my model and the one of HL, and a Wald test for nested models shows that accounting for the share of preference utilisation matters. My model identifies robust uncertainty effects, whereas the HL model finds these effects to be more limited. I ascribe these differences to a misspecification of the

empirical equation in the context of conditional tariff preferences, which introduces measurement error.

A partial equilibrium quantification shows that by the end of 2019, uncertainty reduced the UK's imports from PTA partners by -2.6%. Continuity Agreements brought the reduction in UK imports by the end of 2019 from -2.6% to -2.3%. This result suggests that restoring trade policy credibility after uncertainty arises is not immediate.

In general, the model applies when firms can serve a foreign market under different trade regimes. The regimes considered here have the following characteristics: a) they differ in terms of variable trade costs; b) the firm can choose them; c) accessing the low variable cost regime entails an additional sunk cost. The model applies to any situation where, by making an initial (once and for all) investment, the firm can lower its variable costs, and where uncertainty affects only one regime. For instance, it will apply to a situation like the NAFTA renegotiation, in which the US threatened to end the agreement. It will also apply to the Generalised System of Preferences, for which developed countries grant preferential access to developing ones as long as they do not become too competitive, generating uncertainty about such tariff preferences (Borchert and Di Ubaldo 2020). While uncertainty about the sunk cost to access the preferential regime could be modelled as well, this is left for future research.

This paper builds on the growing literature on trade policy uncertainty. The HL model has been applied extensively, showing how trade policy uncertainty affects the extensive margin of trade (Handley and Limão 2015), the intensive margin (Handley 2014) as well as product prices and quality (Handley and Limão 2017, Feng et al. 2017) and input choices (Handley, Limão et al. 2024). These papers sit on a broader literature on international trade and uncertainty. For instance, Esposito (2022) shows how firms can diversify their exports geographically to reduce their exposure to country-specific risks when trading internationally. Fillat and Garetto (2015) model multinationals' sales strategy when demand is uncertain, and show how sunk entry cost can generate hysteresis, while Ramondo et al. (2013) revisit the proximity-concentration trade-off when demand is subject to country-specific uncertainty shocks.

In the context of Brexit, Crowley et al. (2020) show how tariff uncertainty reduced the number of UK firms trading with the EU. Graziano et al. (2021), who also model variation of uncertainty over time, find that UK-EU trade started to react to Brexit-related uncertainty in the months leading up to the referendum, and Graziano et al. (2024) use the same framework to evaluate the effect of Brexit uncertainty on trade prices, while Ahmad et al. (2023) study uncertainty in services trade. These studies find that Brexit uncertainty negatively affected both the extensive and intensive margin of UK-EU trade, and it increased trade prices. Graziano et al. (2020) look at UK trade with a subset of PTA partners before the 2016 referendum, and to the best of my knowledge, this is the only study that looks at Brexit and the UK's PTA partners. In this case, the authors model uncertainty over time but do not consider tariff preference utilisation rates.

The paper is also linked to the literature on preference utilisation rates. This body of literature generally finds that utilising tariff preferences in trade agreements involves fixed costs. For instance, Keck and Lendle (2012) state that ‘utilization costs are principally of a fixed cost nature’. Similar conclusions are reached by Hayakawa (2015), Hayakawa, Kim et al. (2014), Nilsson and Dotter (2012), Ulloa and Wagner (2012) and Takahashi and Urata (2010).

This paper offers three main contributions. First, I extend the HL model to regimes with conditional tariff preferences and show that the trade policy uncertainty measure of the HL model must be generalised to account for the preference utilisation rate. Given the recent increase in renegotiations of PTAs this can be an important extension of the standard model allowing researchers to study uncertainty in PTAs. In the empirical application, I construct a Wald test for nested models, which rejects the restriction of full exposure to uncertainty. Ignoring information on exposure to the potential tariff increase can lead to different conclusions about the effects of uncertainty on trade flows. Second, I exploit data on preference utilization in a structural model of trade to measure the impact of Brexit uncertainty on the UK’s trade with PTA countries, which is an important policy question. Finally, to estimate the impact of the Continuity Agreements, which have a staggered roll-out, I adapt the flexible regression method of Wooldridge (2021) to a triple-difference estimator.

The rest of the paper is organised as follows. Section 2 presents the policy background on Brexit and UK’s trade with its PTA partners. Section 3 introduces the theoretical model and section 4 explains how I move from theory to the empirical application. Section 5 summarises the data sources and section 6 presents the results. Section 7 concludes.

2 The UK, PTA partners and Brexit

In 2015 the UK had PTAs with 54 countries which accounted for 11% of its imports and 14% of its exports.² These agreements are of different natures. Some are long-lasting agreements such as the one with Switzerland (since 1973) or Turkey (since 1995), while other agreements are more recent.

In June 2016, the UK held a referendum on leaving the EU after forty-three years of membership. Against all expectations, with a thin majority of 52% to 48%, the people voted to leave the European Union. The unexpected result generated a great deal of uncertainty, with the future of UK-EU relations remaining unclear for some time to come. Even more uncertain was the future of the UK’s trade policy beyond the EU. While the official UK exit process from the EU started in March 2017 when the UK invoked Article 50 of the Treaty of the European Union, it was not until the end of 2017 that the UK

²The shares are computed using HMRC trade data for 2015, including UK’s trade with the EU. Excluding trade with the EU, the shares of PTA countries in 2015 are 26% for exports and 25% for imports.

declared its intent to roll over existing PTAs with the Trade Bill 2017-19.³ At the time the bill passed, the deadline for leaving the EU with or without a deal was set for March 2019. By January 2019, three months before the planned departure from the EU, the UK had not finalized any Continuity Agreements. By February’s end, it had only signed CAs with 9 countries, covering 6 of the original 40 EU PTAs (Magntorn Garrett (2019 [Online])). Despite subsequent progress, in 2019 uncertainty regarding the UK’s ability to lock in existing preferences was pervasive.⁴ While some of the large trade partners signed the Continuity Agreement early in 2019 (Switzerland in February; Norway in April), other signed only much later (South Korea in September 2019 and South Africa in October 2019; Turkey in 2021).

The CAs aimed to maintain current preferences and remove uncertainty. As the notes prepared for the UK Parliament by the Department for International Trade (DIT) read, with Continuity Agreements ‘the Government has sought to deliver the maximum possible certainty to businesses and consumers through ensuring continuity in the UK’s existing trade relationships.’ CAs try to replicate the provisions of the existing EU PTAs. Changes can be as simple as replacing ‘EU’ with ‘UK’ in the text of the agreement or more complicated such as adjusting quotas to reflect the size of the UK market.

However, changes are not always so smooth. In terms of Rules of Origins, the cumulation of EU materials and processing is not always allowed by Continuity Agreements, and in some cases, it is only temporarily allowed (see DIT 2019a for UK-South Korea or DIT 2020 for UK-Mexico). Beyond Rules of Origins, some details of EU PTAs might be difficult to replicate. For instance, the EU-Switzerland relation is characterised by a multitude of agreements and it was not possible to replicate some of the existing provisions as they depended on the not yet defined UK-EU trade arrangements (DIT 2019b).

Given that the roll-over started late, it is often incomplete and subject to conditions, it is not obvious whether CAs removed all the uncertainty. While the UK government has sought to deliver ‘the maximum possible certainty’, whether it succeeded is ultimately an empirical question.

3 Theory

This section introduces the theoretical model, which builds on the Melitz (2003) framework. Consumers have CES preferences for differentiated goods, with an elasticity of substitution σ and an outside numeraire good. Firms operate under monopolistic competition, each with a distinct productivity level, denoted by φ , and labour is the only factor of production. Deterministic versions of the model were also used by Demidova and Krishna (2008) and Hayakawa, Naoto et al. (2019). I confine the full description of the background settings to the Appendix.

³See <https://bills.parliament.uk/bills/2172>

⁴To date the UK government signed CAs with 66 PTA partners and only six countries remain uncovered.

Deterministic scenario: An exporter of product h from country i can export to a PTA partner country j under two regimes: the MFN or the preferential (PRF) regime. Under the MFN regime, exporters pay an initial sunk cost f_{hij} and face the MFN tariff τ_{hjt}^{MFN} ($= 1 + \text{MFN tariff}$). Alternatively, exporters can pay an additional sunk cost f_{hij}^{PRF} (hence a total of $f_{hij} + f_{hij}^{PRF}$) and face the preferential tariff $\tau_{hjt}^{PRF} < \tau_{hjt}^{MFN}$ (similarly, $\tau_{hijt}^{PRF} = 1 + \text{PRF tariff}$). The additional sunk cost f_{hij}^{PRF} summarises the fixed cost of accessing preferences. These can be complying with Rules of Origins and dealing with border formalities related to preferential access. These sunk costs are paid only upon entry, and, consistent with the HL model, there are no per-period fixed costs. While I follow the literature on preference utilisation to assume that the costs of utilising PTAs are mainly fixed, the theoretical results are robust to the introduction of additional variable costs of utilising tariff preferences.

Firms are forward-looking and operate with an infinite time horizon and a discount factor $\beta < 1$. A firm chooses to export under the MFN regime if the present value of exporting $V_{hijt}^{MFN} = \frac{\pi_{hijt}^{MFN}(\varphi)}{1-\beta}$ covers the sunk cost f_{hij} . Setting $V_{hijt}^{MFN} = f_{hij}$ and solving for φ we can find the productivity cut-off φ_{hijt}^{MFN} above which firms export under the MFN regime:

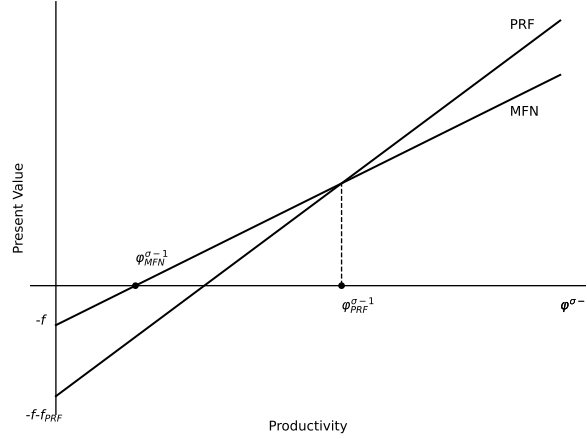
$$\varphi_{hijt}^{MFN} = \left[\frac{f_{hij} (1 - \beta)}{A_{ijt} (\tau_{hjt}^{MFN})^{-\sigma}} \right]^{\frac{1}{\sigma-1}} \quad (1)$$

Trade under the preferential regime involves higher sunk costs and lower variable costs than the MFN regime, hence only more productive firms will select into the preferential regime. This result is the parallel of the exports-FDI nexus of Helpman et al. (2004). The sunk costs of exporting under the PRF regime are given by $f_{hij} + f_{hij}^{PRF}$. To find the cut-off for preferential trade, set $V_{hijt}^{PRF} - V_{hijt}^{MFN} = f_{hij}^{PRF}$, with $V_{hijt}^{PRF} = \frac{\pi_{hijt}^{PRF}(\varphi)}{1-\beta}$, and solve for productivity to find:

$$\varphi_{hijt}^{PRF} = \underbrace{\left[\frac{f_{hij} (1 - \beta)}{A_{ijt} (\tau_{hjt}^{MFN})^{-\sigma}} \right]^{\frac{1}{\sigma-1}}}_{=\varphi_{hijt}^{MFN}} \underbrace{\left[\frac{f_{hij}^{PRF} / f_{hij}}{(\tau_{hjt}^{MFN} / \tau_{hjt}^{PRF})^\sigma - 1} \right]^{\frac{1}{\sigma-1}}}_{=F_{hijt}} \quad (2)$$

that is, the PRF cut-off is equal to the MFN one multiplied by a factor $F_{hijt} = \left[\frac{f_{hij}^{PRF} / f_{hij}}{(\tau_{hjt}^{MFN} / \tau_{hjt}^{PRF})^\sigma - 1} \right]^{\frac{1}{\sigma-1}}$ which summarises the fixed vs variable costs savings between the two regimes. If the proportional increase in sunk cost f_{hij}^{PRF} / f_{hij} is high or if the difference between MFN and PRF tariff is small ($\tau_{hjt}^{MFN} / \tau_{hjt}^{PRF} \sim 1$),

Figure 2: Value functions deterministic model



then the PRF cut-off will be high meaning that only a small fraction of firms will find it profitable to trade under the preferential regime.⁵

To summarise the deterministic model, Figure 2 shows the value functions and the cut-offs of the MFN and PRF export strategies for the scenario in which both MFN and preferential trade are observed.

Uncertainty scenario: When the PTA is uncertain, the exporting firm can choose the preferential regime, but now with uncertainty about future preferential tariffs. The uncertainty is about whether in the next period tariff preferences will remain in place or if these will be revoked and she will have to pay the MFN tariff. Alternatively, she can export under the MFN regime, with no uncertainty about future MFN tariff rates, and switch to the PRF regime when uncertainty resolves. Since MFN tariffs are not uncertain, the MFN entry cut-off is the same as in the deterministic scenario φ_{hijt}^{MFN} .⁶

The availability of the MFN strategy and the complementarity of the initial sunk costs imply that no firm will find it optimal to wait to export. The complementarity of sunk costs means that there is nothing to lose by starting the MFN strategy today. Because waiting gives zero immediate profit, starting under the MFN regime always dominates waiting.⁷

On the other hand, uncertainty about preferential tariffs pushes the PRF cut-off upward, and accessing

⁵In this model, we observe both MFN and PRF trade if the relative fixed and variable costs are within a certain range. The conditions are reported in the Appendix.

⁶This is because the possibility to switch to the PRF regime in the future is not considered by all exporters, but only by those productive enough to choose the PRF regime without uncertainty.

⁷In case the additional fixed costs of utilizing preferences are negligible such that the PRF strategy always dominates the MFN one, then we are back to the standard Handley and Limao model where waiting takes place.

preferences become less attractive. The uncertain PRF regime has a current tariff τ_{hijt}^{PRF} but expected future tariffs $E_t(\tau')$ drawn from a distribution $H(\tau')$ which can vary across industries but it is assumed fixed in time. The present value of exporting under the preferential regime is:

$$V_{hijt}^{PRF} = \pi(\tau_{hijt}^{PRF}) + \beta [(1 - \gamma)V_{hijt}^{PRF} + \gamma E_t V(\tau')] \quad (3)$$

where $\pi(\tau_{hijt}^{PRF})$ is the immediate profit at the current preferential tariff τ_{hijt}^{PRF} , β is a discount factor and γ is the probability of a policy change in $t + 1$ as perceived by the exporter. Equation (3) says that if in the next period nothing changes – which happens with probability $(1 - \gamma)$ – the present value of the PRF regime in $t + 1$ will be the same as today. If it changes, the expected continuation value is $E_t V(\tau') = \frac{E_t \pi(\tau')}{1 - \beta}$. This strategy involves sunk costs $f_{hij} + f_{hij}^{PRF}$.

On the other hand, if the firm starts selling under the MFN regime today she has the option to switch to the preferential regime in case future conditions are favourable, which will occur with probability $H(\bar{\tau})$. Here $\bar{\tau}$ is the future tariff rate below which choosing the PRF regime is optimal. Switching implies paying the additional cost f_{hij}^{PRF} . The present value of the MFN regime for those that consider switching is:

$$V_{hijt}^{MFN} = \pi(\tau_{hijt}^{MFN}) + \beta(1 - \gamma)V_{hijt}^{MFN} + \beta\gamma \left[\underbrace{H(\bar{\tau}) (E_t V(\tau' | \tau' \leq \bar{\tau}) - f_{hij}^{PRF})}_{\text{switch to PRF}} + \underbrace{(1 - H(\bar{\tau})) V_{hijt}^{MFN}}_{\text{continue under MFN}} \right] \quad (4)$$

where $\pi(\tau_{hijt}^{MFN})$ is the immediate profit at the current MFN tariff. Tomorrow, with probability $1 - \gamma$ nothing will change so the present value remains the same. With probability $\gamma H(\bar{\tau})$, where $H(\bar{\tau})$ is the probability that $\tau' \leq \bar{\tau}$, the policy change is favourable and the firm switches to the preferential regime after paying the extra cost f_{hij}^{PRF} . Finally, with probability $\gamma(1 - H(\bar{\tau}))$ the policy will change but the firm will still prefer the MFN to the PRF regime. This is because the new tariff τ' is above the threshold $\bar{\tau}$ below which the firm would prefer the PRF to the MFN regime.

The probability of a policy change γ is taken as given by the firm. However, this probability can be influenced with credible policies. For instance, a credible commitment to the PTA will drive γ to zero (Carballo et al. 2022).

To find the productivity cut-off above which the firm will chose the PRF regime under uncertainty (call this regime UPRF), set the difference in present values equal to the difference in sunk costs $V_{hijt}^{PRF} -$

$V_{hijt}^{MFN} = f_{hij}^{PRF} + f_{hij} - f_{hij}$ and solve for productivity to get the cut-off under the UPRF regime φ_{hijt}^{UPRF}

$$\varphi_{hijt}^{UPRF} = \underbrace{\left[\frac{f_{hij}(1-\beta)}{A_{ijt}(\tau_{hijt}^{MFN})^{-\sigma}} \right]^{\frac{1}{\sigma-1}} \left[\frac{f_{hij}^{PRF}/f_{hij}}{(\tau_{hijt}^{MFN}/\tau_{hijt}^{PRF})^\sigma - 1} \right]^{\frac{1}{\sigma-1}}}_{=\varphi_{hijt}^{PRF}} \underbrace{\left[\frac{1-\beta+\beta\gamma}{1-\beta+\beta\gamma\omega_{hijt}^{PRF}} \right]^{\frac{1}{\sigma-1}}}_{=U_{hijt}} \quad (5)$$

with

$$\omega_{hijt}^{PRF} = 1 - (1 - H(\tau_{hijt}^{PRF})) \frac{(\tau_{hijt}^{PRF})^{-\sigma} - E_t(\tau^{-\sigma}|\tau' > \tau_{hijt}^{PRF})}{(\tau_{hijt}^{PRF})^{-\sigma} - (\tau_{hijt}^{MFN})^{-\sigma}} < 1 \quad (6)$$

where I replaced $\bar{\tau} = \tau_{hijt}^{PRF}$. As noted in Handley and Limão (2015), this substitution involves asking what is the productivity cut-off φ^{UPRF} above which a firm would enter the PRF regime given the current PRF tariff rate. The inequality in (6) follows from the fact that $E_t(\tau^{-\sigma}|\tau' > \tau_{hijt}^{PRF}) < (\tau_{hijt}^{PRF})^{-\sigma}$ as $\sigma > 1$.

The PRF cut-off under uncertainty in (5) is the deterministic cutoff φ_{hijt}^{PRF} multiplied by the term $U_{hijt} = \left[\frac{1-\beta+\beta\gamma}{1-\beta+\beta\gamma\omega_{hijt}^{PRF}} \right]^{\frac{1}{\sigma-1}}$. The key result that $\omega_{hijt}^{PRF} < 1$ (proof in the Appendix) implies that $U_{hijt} > 1$ and therefore that the PRF cut-off under uncertainty is larger than the deterministic one, shifted to the right by U_{hijt} . Note that in case of no uncertainty ($\gamma = 0$) we have $U_{hijt} = 1$ and therefore the PRF entry cut-off is just the deterministic one.

The term ω_{hijt}^{PRF} depends on the probability of a bad draw $1 - H(\tau_{hijt}^{PRF})$, and on the tariff margin between the preferential and the expected higher tariff $E_t(\tau^{-\sigma}|\tau' > \tau_{hijt}^{PRF})$ relative to the gap between the preferential and MFN tariff. When the firm expects that losing preferential tariffs would result in reversion to MFN, such that $E_t(\tau^{-\sigma}|\tau' > \tau_{hijt}^{PRF}) = (\tau_{hijt}^{MFN})^{-\sigma}$, the shift in the preferential cut-off due to uncertainty becomes independent of the tariff margin. Intuitively, the deterministic cut-off already accounts for how much better the preferential regime is relative to MFN, so this information is already embedded in the firm's initial decision. Under uncertainty, what matters is the probability that preferential access will be lost. Nonetheless, changes at the intensive margin – the value of trade – will depend on the PRF/MFN tariff margin, as explained in the next section.

Export value of new firms: We can find the value of exports of firms entering in period t by integrating firm sales over productivity. In the deterministic case, total exports consist of two parts: firms that export under the MFN regime (with productivity between φ_{hijt}^{MFN} and φ_{hijt}^{PRF}) and firms that choose the preferential regime (with productivity above φ_{hijt}^{PRF}). Under uncertainty, we decompose total exports into three components: (i) firms that export under MFN regardless of uncertainty ('always MFN'); (ii) firms

that would have used the PRF regime but switch to MFN due to uncertainty ('switchers'); and (iii) firms that use the PRF regime with or without uncertainty ('always PRF'). Total exports under uncertainty are then given by:

$$X_{hijt} = \underbrace{\int_{\varphi_{hijt}^{MFN}}^{\varphi_{hijt}^{PRF}} \pi_{hijt}^{MFN} dG(\varphi)}_{\text{always MFN}} + \underbrace{\int_{\varphi_{hijt}^{PRF}}^{\varphi_{hijt}^{UPRF}} \pi_{hijt}^{MFN} dG(\varphi)}_{\text{switchers}} + \underbrace{\int_{\varphi_{hijt}^{UPRF}}^{\infty} \pi_{hijt}^{PRF} dG(\varphi)}_{\text{always PRF}} \quad (7)$$

Uncertainty does not affect the 'always MFN' firms directly, as they never find it profitable to use preferences. 'Always PRF' firms continue to use the preferential regime, despite higher fixed costs, due to their high productivity.⁸ However, the key adjustment occurs via the switchers: these firms would use the PRF regime in a deterministic environment, but under uncertainty they prefer to delay and export under MFN. This leads to lower export values, since MFN tariffs are higher than PRF tariffs.

Crucially, this intensive margin effect operates even if no firms exit – in contrast to the HL model where uncertainty affects the extensive margin by reducing the number of exporters. In this model, the mass of exporters is unchanged, but a growing share of firms exports under MFN, reducing the aggregate value of trade. Overall, the effect of uncertainty on trade values passes via two channels. At the extensive margin, uncertainty pushes more firms into the MFN regime. As these firms face the MFN instead of the preferential tariff, they export less than they would have done in the absence of uncertainty.

To better understand how policy uncertainty affects the value of trade, we can look at the semi-elasticity of imports with respect to the probability of a policy change γ . First, I parametrise the firm distribution using a Pareto distribution as customary in the trade literature with a Cumulative Density Function $G(\varphi) = 1 - \varphi^{-\alpha}$ and shape parameter α . As shown in the Appendix, total imports of new entrants can be expressed as:

$$X_{hijt} = C_{hij} B_{jt} m_{it} \left(\tau_{hijt}^{MFN} \right)^{-\frac{\sigma\alpha}{\sigma-1}} \times \left[1 + \left(\left(\frac{\tau_{hijt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^{\sigma} - 1 \right) (F_{hijt} \times U_t)^{-\alpha+\sigma-1} \right] \quad (8)$$

where C_{hij} , B_{jt} summarise aggregate conditions at the exporter-importer-product and importer-time, respectively, while m_{it} is the mass of firms in the exporting country. The semi-elasticity of imports with

⁸As uncertainty affects the price index, there is an aggregate change in exports of the 'always MFN' firms. However, this price index effect applies also to the 'switchers' and 'always PRF' groups, hence we consider it an aggregate demand shifter.

respect to γ is given by:

$$\begin{aligned} \frac{\partial \ln X_{hijt}}{\partial \gamma} = & \underbrace{-\frac{\alpha - \sigma + 1}{\sigma - 1}}_{\text{extensive margin elasticity}} \times \underbrace{\frac{\beta(1 - \beta)}{(1 - \beta + \beta\gamma)(1 - \beta + \beta\gamma\omega_{hijt}^{PRF})}}_{\text{discounted exp. value adjustment}} \times \underbrace{(1 - \lambda)}_{\text{probability of bad draw}} \times \\ & \underbrace{\frac{(\tau_{hijt}^{MFN}/\tau_{hijt}^{PRF})^\sigma (F_{hijt} \times U_t)^{-\alpha + \sigma - 1}}{1 + \left[(\tau_{hijt}^{MFN}/\tau_{hijt}^{PRF})^\sigma - 1 \right] (F_{hijt} \times U_t)^{-\alpha + \sigma - 1}}}_{\text{share of preferential imports}} \times \underbrace{\left[1 - \frac{E_t(\tau^{-\sigma} | \tau' > \tau_{hijt}^{PRF})}{(\tau_{hijt}^{PRF})^{-\sigma}} \right]}_{\text{potential profit loss}} < 0 \end{aligned} \quad (9)$$

The semi-elasticity in (9) is composed of five elements. The term $\frac{\alpha - \sigma + 1}{\sigma - 1}$ is the extensive margin elasticity $(\alpha - \sigma + 1)$ as in Chaney (2008), scaled by the love of variety term $(\sigma - 1)$. The second term discounts the future, accounting for the probability of a policy change, while $(1 - \lambda)$ is the probability of a bad draw in case of a policy change – i.e., that the new tariff will be above the current preferential one – which is assumed to be common across all products. The last two terms complete the measurement of risk. The first term in the second line of (9) is the share of imports coming under the preferential regime, measuring the exposure to the risk of losing preferential tariffs. The last term is the potential profit loss, or how much a firm could lose if the new tariff will be above the current preferential rate.

Effectively, the last two terms of equation (9) represent the empirical measure of uncertainty. In my application to Brexit, where if preferential tariffs are lost trade would revert to the MFN regime, we can replace $E_t(\tau^{-\sigma} | \tau' > \tau_{hijt}^{PRF})$ with the MFN tariff $(\tau_{hijt}^{MFN})^{-\sigma}$. Hence, the risk of losing preferential tariffs for the MFN regime will be measured by multiplying the share of preferential imports s_{hijt}^{PRF} by the potential profit loss $\omega_{hijt} = \left[1 - (\tau_{hijt}^{PRF}/\tau_{hijt}^{MFN})^\sigma \right]$.

4 Empirical approach

This section derives an empirical model for trade values that accounts for regime heterogeneity, forward-looking uncertainty, and the gradual adjustment of the extensive margin through firm entry. The key contribution is to operationalize the effects of policy risk on aggregate trade flows, using observable data on tariff preferences and preference utilization.

Uncertainty influences trade values primarily through the firm's extensive margin decision of whether to operate under the MFN or preferential trade regime. In line with Graziano et al. (2021), uncertainty impacts the trade regime choices of new entrants exclusively. Firms that began trading before an uncertainty shock do not change their trade regime in response to this shock. This implies that the full effect of an uncertainty shock on trade values takes time to manifest, as it depends on the gradual adjustment

of new firms entering the market.

Accounting for past entry conditions, total exports of product h from i to j in period t can be written as:

$$X_{hijt} = B_{jt} C_{hij} \left(\tau_{hjt}^{MFN} \right)^{-\sigma} \times \left\{ \sum_{l=0}^{\infty} (1-\psi)^l m_{it-l} \left(B_{jt-l} \left(\tau_{hjt-l}^{MFN} \right)^{-\sigma} \right)^{\frac{\alpha-\sigma+1}{\sigma-1}} \left[1 + \left(\left(\frac{\tau_{hjt}^{MFN}}{\tau_{hjt}^{PRF}} \right)^{\sigma} - 1 \right) (F_{hijt-l} U_{t-l})^{-\alpha+\sigma-1} \right] \right\} \quad (10)$$

The term in curly brackets in (10) represents the current entry conditions at time t as well as past entry conditions of time $t-l$. The latter are discounted by the exogenous exit shock ψ , with $(1-\psi)^l$ representing the survival rate after l periods.

I derive the empirical equation by taking a log-linear approximation of (10) of all time-varying quantities around a steady state equilibrium $t = 0$ with no uncertainty $\gamma = 0$ (see the Appendix for details). Dropping the importer subscript j given that the UK will be the only importer in the dataset, I obtain the following expression:

$$\begin{aligned} \ln X_{hit} = & \underbrace{a_{hi} + a_{it}}_{\text{fixed effects}} - \left(\sigma + \sigma \frac{\alpha - \sigma + 1}{\sigma - 1} \psi \right) \times \underbrace{\left[s_{hi}^{MFN} \ln \tau_{ht}^{MFN} + s_{hi}^{PRF} \ln \tau_{hit}^{PRF} \right]}_{\text{current applied average tariff}} + \\ & - \sum_{l=1}^{\infty} \left[\sigma \frac{\alpha - \sigma + 1}{\sigma - 1} \psi (1-\psi)^l \right] \times \underbrace{\left[s_{hi}^{MFN} \ln \tau_{ht-l}^{MFN} + s_{hi}^{PRF} \ln \tau_{hit-l}^{PRF} \right]}_{\text{lagged applied average tariff}} + \\ & - \underbrace{\frac{\alpha - \sigma + 1}{\sigma - 1} \frac{\psi \beta}{1 - \beta} (1 - \lambda) \left[\sum_{l=0}^{\infty} (1-\psi)^l \gamma_{t-l} \right]}_{\text{uncertainty coefficient}} \times \underbrace{\left(s_{hi}^{PRF} \times \omega_{hi} \right)}_{\text{uncertainty measure}} \quad (11) \end{aligned}$$

The first two terms summarise aggregate conditions at the product-by-partner (a_{hi}) and partner-by-time (a_{it}) level, and can be absorbed into fixed effects. s_{hi}^{PRF} is the equilibrium (pre-uncertainty) share of PRF trade of product h from country i and $s_{hi}^{MFN} = 1 - s_{hi}^{PRF}$ is the MFN share.

The modified gravity equation (11) shows that for PTA countries the applied tariff should be the log of the geometric average of the MFN and PRF tariff, with weights given by the preferential import share. The equation includes lags of the applied tariff in line 2 to account for past entry conditions.

The last line in (11) measures the effect of uncertainty on trade. The risk of an event occurring can be measured as the product of the probability of the event, the exposure to the event and the potential loss. The last line of (11) includes all these three elements. The potential loss, also called tail risk in Graziano et al. (2021), is measured by $\omega_{hi} = 1 - (\tau_{hi}^{PRF} / \tau_h^{MFN})^{\sigma}$. I assume $\sigma = 4$, in line with the trade literature. This is the loss in profits that a firm would face in case preferential tariffs are lost and trade reverts to the

MFN regime. It is evaluated before the uncertainty shock.⁹

The exposure to the risk of losing preferential tariffs is measured with the share of preferential trade s_{hi}^{PRF} . This tells us that, if tariff preferences are not utilised, uncertainty will have no effect on trade. In equation (11), uncertainty will be effectively measured by $(s_{hi}^{PRF} \times \omega_{hi})$, the share of preferential trade multiplied by the potential profit loss.

Finally, the probability of the event occurring is embedded within the coefficient to be estimated on the uncertainty term. This coefficient includes the history of uncertainty shocks $\left[\sum_{l=0}^{\infty} (1 - \psi)^l \gamma_{t-l}\right]$, which can vary over time. Consider the arrival of a lasting uncertainty shock in period t . Initially, only the exports of new entrants in period t are affected. In the second period, both the exports of the new entrants in $t + 1$ and surviving firms that entered in t are affected by uncertainty. Over time, more waves of new firms will respond to the uncertainty, while some firms will have exited. Consequently, the history of uncertainty shocks is discounted by the exit probability ψ .

4.1 Taking the model to the data

The 2016 Brexit referendum presents a unique context for examining trade policy uncertainty. When the UK voted to leave the EU, the outcome of the referendum was largely unexpected, and the result was a tight 52% to 48% for leaving the EU. Moreover, as Brexit was related to UK-EU relations, PTA countries were just third parties in the referendum. Hence, for the UK PTA partners Brexit represented a largely unanticipated exogenous shock that can be used as a natural experiment.

To estimate the effects of Brexit uncertainty on UK trade with its PTA partners, I use monthly imports of the UK from its PTA partners over 2013-19. The period 2013-15 is the base period with no uncertainty. Uncertainty is introduced by the Brexit referendum which occurred in June 2016. In 2019 some countries might have experienced reduced uncertainty as they signed the Continuity Agreement.

Uncertainty effects are estimated by comparing 2013-15 to 2016-19 across products and countries with different levels of risk. To account for extensive margin dynamics, I estimate uncertainty coefficients for each month over 2016-19.

In terms of uncertainty, the treated units are the country-product pairs with a positive share of preferential imports and a positive gap between the MFN and PRF tariffs. The uncertainty treatment is a continuous measure.

The effect of Continuity Agreements is estimated by comparing products exposed to different levels of uncertainty before and after the signature of the CA and across countries. If the CA successfully eliminated uncertainty, the difference between products exposed to uncertainty and those that were not

⁹This is different from expression (6). In the standard Handley and Limao model, uncertainty is measured only with ω_{hi} .

should disappear after the agreement is signed. Comparison are made over time, across products with different tariff risks, and across countries that signed or not the CA. The cross-country comparison allows me to control for the evolving effects of uncertainty over time in the estimation of the CA's impact.

Given the possible dynamic effects of uncertainty, the staggered adoption of Continuity Agreements by different countries might create identification issues for CA effects (Goodman-Bacon 2021). To correct for this potential issue, I adapt the approach of Wooldridge (2021) to a triple-difference estimator. This implies estimating cohort-time average treatment effects (ATT), where a cohort is a group of countries that signed the CA in the same month of 2019 (e.g., for the February cohort, I estimate ATTs for February-December 2019). The control group is defined as the set of countries not subject to treatment (the CA) in the period for which the ATT is computed. Countries that did not sign a CA in 2019 are always part of the control group, while countries that signed CAs in 2019 leave the control group from the month in which they signed it.

The econometric model to be estimated with monthly data by linear regression is:

$$\ln X_{hit} = a_{hi,month} + a_{it} + a_{ht} + \sum_{l=0}^3 b_l \overline{\ln \tau_{hit-l}} + \sum_{t=Jan2016}^{Dec2019} b_t^\gamma \left(s_{hi}^{PRF} \times \omega_{hi} \times a_t \right) + \sum_c \sum_{t \geq t_{CA}} b_{c,t}^{CA} \left(CA_{ct} \times a_t \times \omega_{hi} \times s_{hi}^{PRF} \right) + e_{hit} \quad (12)$$

where t indicate a month-year (e.g., January 2015). The $a_{..}$ are fixed effects at the product-country-calendar month, country-time and product-time level. The subscript c indicates a Continuity Agreement signatory cohort – e.g., all countries that signed in February 2019.

The term $\overline{\ln \tau_{hit-l}} = (1 - s_{hi}^{PRF}) \ln \tau_{hit-l}^{MFN} + s_{hi}^{PRF} \ln \tau_{hit-l}^{PRF}$ is the applied tariff. I include up to three lags of the tariffs (yearly frequency), and estimate separate coefficients for the current tariff and each of the lagged tariffs. Including more lags means dealing with more changes in product classification over time which might introduce some bias. Moreover, there is only little variation in tariffs over time in the sample.

The baseline PRF shares s_{hi}^{PRF} and the potential profit loss ω_{hi} are computed with data for 2013-15. For the PRF share, this means that if a partner-product observation was not traded in 2013-15 but it was traded in 2016-19, it is dropped from the sample. This affects around 6% of observations. As a robustness test, I compute PRF shares for the entire period 2013-19 and instrument them with the EU27 shares for the same period. I also provide a robustness test instrumenting ω_{hi} with MFN tariffs of other advanced economies.

I estimate uncertainty effects for each month between January 2016 and December 2019 by interacting

the product-level measure of uncertainty $s_{hi}^{PRF} \times \omega_{hi}$ with time dummies a_t . Each coefficient b_t^γ measures the cumulative effect of uncertainty shocks up to period t . As a robustness test, I also use a time-varying index of Brexit uncertainty in a specification similar to Graziano et al. (2021).

The cohort-time CA effects are measured by the coefficients $b_{c,t}^{CA}$. The dummy CA_{ct} takes the value of one if country i in cohort c signed a CA in period t and zero otherwise. For the HL model, the estimating equation is derived under the restriction $s_{hi}^{PRF} = 1$. To test whether such restriction is innocuous, in the next section I introduce a Wald test for nested models.

4.2 A Wald test for nested models

If we were to study the effect of uncertainty about preferential tariffs with the standard HL model (e.g., as done by Graziano et al. 2020), we would treat the preferential regime as the only available regime before uncertainty. We can derive the HL empirical equation by setting $s_{hi}^{PRF} = 1$ in equation (12).

As this restriction is on data rather than coefficients, equation (12) nests the HL regression model *mathematically* but not *econometrically*. However, a simple variable substitution yields an econometrically nested structure. This involves substituting the share of preferential imports s_{hi}^{PRF} with $(1 - s_{hi}^{MFN})$ in (12) and rearranging terms. For the purpose of model testing, I exclude the effects of Continuity Agreements, I interact the uncertainty measure with a post-referendum dummy and include only the current tariff.¹⁰ We then have:

$$\begin{aligned} \ln M_{hit} = & a_{hi} + a_{it} + a_{ht} + \underbrace{b_\tau^{HL} \ln \tau_{hit}^{PRF} + b_U^{HL} (\omega_{hi} \times Post_t)}_{\text{HL model}} + \\ & \underbrace{b_\tau^{PRF} [s_{hi}^{MFN} \times \ln (\tau_{ht}^{MFN} / \tau_{hit}^{PRF})] + b_U^{PRF} (s_{hi}^{MFN} \times \omega_{hi} \times Post_t)}_{\text{Additional terms}} + e_{hit} \end{aligned} \quad (13)$$

The HL model is represented in the second line. This includes the preferential tariff $\ln \tau_{hit}^{PRF}$ as well as the tail risk, or potential profit loss, ω_{hi} interacted with the year dummies post-referendum dummy $Post_t$. The terms in the third line are the additional terms of my model. A test on the joint significance of b_τ^{PRF} and b_U^{PRF} is a test on the nested structure of the model. If we reject the null hypothesis, the data reject the restriction imposed by the HL model in favour of the model presented in this paper.

Note that the in (13) we expect the uncertainty coefficient of the HL model b_U^{HL} to be negative, while, because of the transformation, the coefficients b_U^{PRF} are expected to be positive. We should also have

¹⁰This is to avoid reporting many coefficients which would make reading the test more difficult. In unreported results I checked that including interactions of uncertainty with year-month dummies and including lagged tariffs does not affect the results of the test.

$$b_{\tau}^{HL} = b_{\tau}^{PRF}.$$

5 Data

I use data on monthly product-level imports of the UK and preference utilisation rates (from Eurostat Comext) and MFN and preferential tariffs (from ITC MacMap). Information on PTAs and Continuity Agreements is taken from the European Union Commission and the UK Government websites, respectively. A more detailed description of the data is reported in the Appendix.

The dataset includes all EU PTA partners which enjoyed preferential access in 2015 to the EU.¹¹ Some of these countries had preferential access to the EU before 2016 under the GSP or EBA schemes, and signed a trade agreement only after 2015, while other countries such as South Africa changed the type of agreement in the sample period. I consider both the full sample of 70 countries and the restricted sample of countries with consistent PTAs in place over 2013-19. This restricted sample counts 41 countries. A full list of PTA partners is reported in Tables 5-6 in the Appendix.

I exclude imports for inward and outward processing as products imported under these regimes are fully or partially exempt from customs duties, hence they are not subject to tariff uncertainty. In the sample, processing trade accounts for 3.6% of total imports in 2015. Empirical results are robust to the inclusion of processing trade.

Table 1 reports the summary statistics of the log of imports (excluding processing trade), PRF and applied tariff rates, and for the other independent variables at the reference period 2013-15. The average share of PRF imports in 2013-15 is 0.52 for the UK and 0.54 for the EU27.

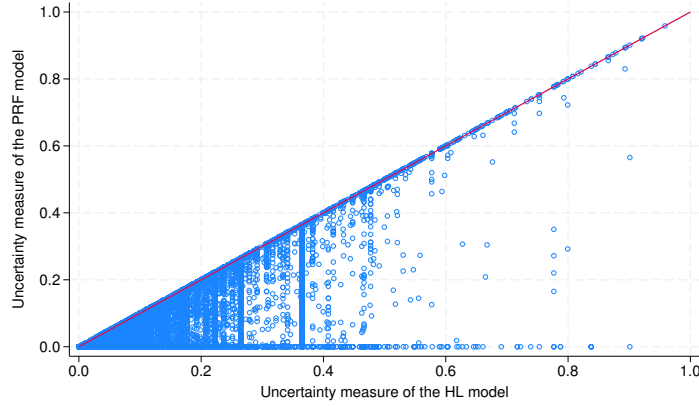
The correlation between the uncertainty measures of the two models, ω_{hi} and $\omega_{hi} \times s_{hi}^{PRF}$, is 0.69 for the observations with non-zero preferential margin. However, for 54% of the partner-CN8 cross-sectional observations with a non-zero preferential tariff margin, the PRF share equals zero. Figure 3 plots the uncertainty measures of the PRF and HL models against each other for each country-CN8 observation. On the diagonal we have the observations for which the share of PRF trade is 1 (about 28% of the cross-sectional dimension), while below the diagonal there are points for which $s_{hi}^{PRF} < 1$ and on the horizontal axis lie the observations with $s_{hi}^{PRF} = 0$. The HL measure would overstate the level of uncertainty for all observations with preferential trade share strictly smaller than one.

¹¹Of the current EU PTA partners, I exclude Canada, Japan, Singapore and Vietnam as they signed trade agreements with the EU after 2015.

Table 1: Summary statistics for UK imports ex. processing trade

	Mean	sd	Median	Min	Max	Count
Log UK imports ($\ln X_{hit,UK}$)	10.018	2.010	9.848	0.000	22.278	849,407
Log EU imports ($\ln X_{hit,EU}$)	11.750	2.776	12.052	0.000	21.444	728,516
Log preferential tariff ($\ln \tau_{hit}^{PRF}$)	0.002	0.020	0.000	0.000	0.776	845,350
Log applied tariff ($\ln \overline{\tau_{hit}}$)	0.012	0.029	0.001	0.000	0.776	794,306
PRF share UK 2013-15 ($s_{hi,UK}^{PRF}$)	0.522	0.449	0.668	0.000	1.000	805,064
PRF share UK 2013-19 ($s_{hi,UK}^{PRF}$)	0.509	0.439	0.599	0.000	1.000	849,335
PRF share EU 2013-19 ($s_{hi,EU}^{PRF}$)	0.540	0.419	0.718	0.000	1.000	837,793
Potential profit loss (ω_{hi})	0.145	0.138	0.101	0.000	0.975	837,683
Potential profit loss IV (ω_{hi}^{IV})	0.097	0.108	0.060	0.000	0.558	845,020
MFN risk = $s_{hi,UK}^{PRF} \times \omega_{hi}$	0.110	0.135	0.060	0.000	0.958	794,316
log Brexit Uncertainty Index	0.245	0.204	0.346	0.000	0.548	849,407

Figure 3: Comparison of uncertainty measures, UK imports



The figure plots the partner-product uncertainty measure of the PRF model against the one of the HL model. In the HL model uncertainty is defined as $\omega = 1 - (\tau^{PRF}/\tau^{MFN})^\sigma$ while in the PRF model this measure is multiplied by the PRF share $\omega \times s^{PRF}$.

6 Results

This section presents the econometric results. First, I apply the Wald test for nested models described in Section 4.2 to see whether the PRF import shares carries additional information. I then move to the estimation of the model including dynamic effects of uncertainty and the effects of the continuity agreement.

Table 2: Wald test

	(1)	(2)	(3)	(4)
$\ln \tau_{hit}^{PRF}$	-1.232*** (0.435)	-1.457*** (0.452)	-2.268*** (0.762)	-3.013*** (0.907)
$Referendum_t \times \omega_{hi}$	-0.194*** (0.048)	-0.313*** (0.051)	0.099 (0.230)	0.050 (0.232)
$s_{hi}^{MFN} \times \ln(\tau_{ht}^{MFN} / \tau_{hit}^{PRF})$		-1.648 (1.007)		-3.044 (1.858)
$Referendum_t \times s_{hi}^{MFN} \times \omega_{hi}$		0.699*** (0.097)		0.992*** (0.136)
Observations	786,768	786,768	636,365	636,365
Exporter-product FE	Yes	Yes	Yes	Yes
Exporter-time FE	Yes	Yes	Yes	Yes
Product-time FE	No	No	Yes	Yes
Wald statistic		54.7		56.3
p-value		0.00		0.00

Standard errors clustered at exporter-product level in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Columns 1 and 3 report estimates of the HL model while columns 2 and 4 report the results for the unrestricted model. The bottom of the table reports the Wald statistic for the test of joint significance of the additional coefficients estimated in the unrestricted model. Columns 1-2 do not include product-time fixed effects, while columns 3-4 do include them. The Wald statistic is distributed χ^2 2 degrees of freedom.

Wald test: Table 2 reports the results for the Wald test for nested models using equation (13). Columns 1 and 3 report the results for the restricted model, while columns 2 and 4 report those for the unrestricted model. The Wald statistic on the joint significance of the additional parameters estimated in the unrestricted model together with its p -value are reported at the bottom of the table.

Columns 1-2 include partner-product and partner-time fixed effects, while columns 3-4 also include product-time fixed effects. Differences in the number of observations are due to the unbalanced nature of the panel dataset and singleton observations being removed (Correia 2016).

The additional uncertainty term in the unrestricted model has the expected positive sign and it is statistically significant. In terms of the Wald test, whether product-time fixed effects are included or not, the data speak clearly. The test statistic is well above conventional critical values, and the restriction imposed by the standard model is rejected. The results of the Wald test tell us that when tariffs are not granted unconditionally, as in the case of a free trade agreement, the share of preference utilisation carries meaningful information.

The coefficient on the interaction of ω_{hi} with the post-referendum dummy is negative and significant in

columns 1-2, as expected. However, when I include product-time fixed effects in columns 3-4 this coefficient turns positive, close to zero and are estimated very imprecisely. On the other hand, the coefficient on the interaction $\omega_{hi} \times s_{hi}^{MFN}$ remains similar in size and significance with and without the product-time FE (compare columns 2 and 4). One possible explanation is that the response of imports to the tail risk ω_{hi} is actually driven by aggregate product-level shocks, hence absorbed by the product-time fixed effects. A second explanation is lack of variation in tariff margins across countries for the same CN8 product.

A decomposition of the variation in ω_{hi} and s_{hi}^{PRF} along countries and products shows that for ω_{hi} most of the variation is between rather than within products. Differently, s_{hi}^{PRF} exhibits similar variation both within and between products (see Table 12 in the Appendix). Hence, product-time fixed effects likely remove much of the variation necessary to estimate the coefficients on ω_{hi} , but at the same time their exclusion might introduce some bias as product-specific shocks are not accounted for. Given that the variable of interest in this paper is the interaction $\omega_{hi} \times s_{hi}^{PRF}$, the rest of the analysis includes product-time FE. I discuss results with different levels of product-time FE in section 6.1.

The coefficients on the preferential tariff and the interaction of the MFN share with the tariff margin are virtually identical, although the latter is insignificant. A test for the equality of the coefficients fails to reject the null hypothesis that the coefficients are the same, in line with the model prediction.

We know for a fact that preference utilisation rates are not always 100%. The econometric test carried out here tells us something more: the information contained in the preference utilisation data is not to be overlooked. This information can help us to understand the response of trade given a shock to trade costs or a change in uncertainty regarding tariff preferences. Data on preference utilisation is likely to be relevant for any analysis of conditional tariff preferences and not for uncertainty only. Given the availability of preference utilisation data, at least for large traders such as the EU and the US, it is somehow surprising that structural models of trade often ignore the conditionality of PTA tariff preferences.

Baseline results: I now move to the estimation of the model with dynamic uncertainty effects and Continuity Agreement effects. Table 3 reports the baseline results. In columns 1-2 I include only the current tariff and the interaction of the uncertainty measure with time dummies from January 2016 to December 2019. In columns 3-4 I add the interaction of the uncertainty measures with the Continuity Agreement and time dummies (February-December 2019), without dealing with the staggered adoption of Continuity Agreements. Finally, columns 5-6 report the full specification with lagged tariffs and Continuity Agreement effects estimated by cohort-period. For each specification I report the estimation of the PRF and the HL models.

The coefficients on the current tariffs are negative and significant in all specifications. Coefficients on lagged tariffs are estimated with little precision, most likely because of the little variation in tariffs over

time in the sample.

The coefficients on uncertainty are estimated for each period between January 2016 and December 2019. For a matter of space I report year averages in Table 3 and plot the individual coefficients in Figure 4 for the PRF model of column 6. For CA effects, I report the average across the monthly effects in columns 3-4, and the average across all cohort-time coefficients in columns 5-6. The disaggregated CA effects are reported in Figure 8 in the Appendix.

The HL model does not reveal evidence of uncertainty affecting UK imports from PTA partners between Jan2016-Dec2019. In contrast, with the PRF model I find negative and significant effects of uncertainty which are increasing in magnitude over time. Ignoring the information contained in the PRF import share leads to measurement error and biased coefficient estimates. From Table 3, we would conclude that there was no effect of uncertainty on UK imports if we ignored the PRF import share, while adding this information to the analysis reverts the conclusions. This difference is due to a misspecification of the model when tariff preferences are not fully utilised.

For the Continuity Agreement effects of columns 5-6, I compute both the simple average of the cohort-time effects, as well as a weighted average, with weights given by number of observations of each cohort. The weighted average is robust to possible erratic behaviour in the simple average due to small cohorts, and it is the preferred measure of CA effect. Both the simple and weighted averages of the cohort-time CA effects are positive and significant for the PRF model. For the HL model, the weighted average effect is positive but the simple average is negative and significant. Examination of the disaggregated cohort-time effects reveals that this negative simple average is driven by a few observations for small countries.¹² For these observations the share of PRF imports is zero, hence the PRF model assigns no uncertainty to them and they are excluded from the calculation of CA effects. Excluding the cohort-time effects for these countries in the HL model yields CA coefficients similar to those of the PRF model.

As the first Continuity Agreement was signed in February 2019, I compare the CA effects with the uncertainty effects over the period February-December 2019. In the PRF model, the average CA effects are positive but smaller than the uncertainty effects in 2019. This suggests that the CA removed only part of the uncertainty that was introduced by the 2016 referendum. In the last panel of Table 3 I report the results for the test that the average CA effect plus the average uncertainty effects in 2019 equals zero. This hypothesis is rejected, and we can see that the CA removed around 30% of the referendum uncertainty.¹³

Figure 4 plots the monthly uncertainty coefficients of the PRF model. There is no evident sign of uncertainty effects before the referendum and a sudden drop in imports just after the referendum in July

¹²The countries are Antigua and Barbuda, the Bahamas and Kosovo.

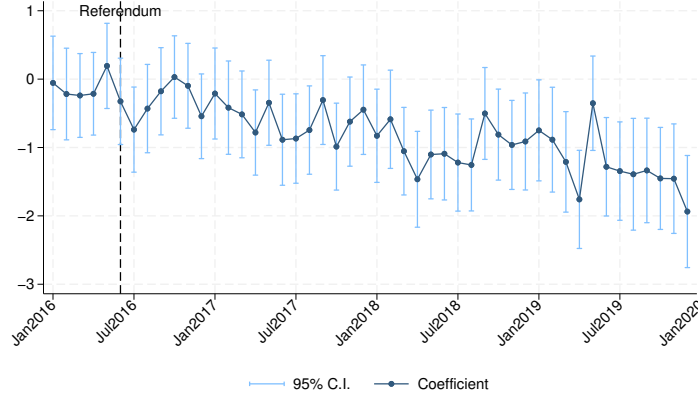
¹³For the test, the average uncertainty effect in 2019 is computed over the period February-December as the first Continuity Agreements were signed in February 2019.

Table 3: Baseline results

	Model 1		Model 2		Model 3	
	(1)	(2)	(3)	(4)	(5)	(6)
	HL	PRF	HL	PRF	HL	PRF
<i>Tariff</i>						
current	-1.827** (0.796)	-1.961* (1.016)	-1.807** (0.795)	-1.969* (1.012)	-1.464* (0.770)	-1.770* (0.979)
1-year lag					-1.416** (0.643)	-1.156 (0.739)
2-year lag					0.136 (0.644)	0.034 (0.708)
3-year lag					-0.123 (0.575)	0.050 (0.648)
<i>Uncertainty effects, yearly average</i>						
2016	0.112 (0.222)	-0.242* (0.143)	0.109 (0.222)	-0.244* (0.143)	0.169 (0.227)	-0.235 (0.144)
2017	0.401 (0.286)	-0.607*** (0.172)	0.395 (0.286)	-0.610*** (0.172)	0.461 (0.292)	-0.594*** (0.175)
2018	0.027 (0.337)	-0.998*** (0.184)	0.023 (0.337)	-1.000*** (0.184)	0.112 (0.353)	-0.983*** (0.187)
2019	0.063 (0.422)	-1.094*** (0.198)	-0.166 (0.433)	-1.276*** (0.207)	-0.084 (0.445)	-1.263*** (0.210)
<i>Continuity Agreements, average Feb-Dec 2019:</i>						
Simple average			0.336** (0.142)	0.327** (0.153)	-0.882*** (0.302)	0.333** (0.167)
Weighted average					0.450*** (0.147)	0.395** (0.162)
<i>Test if Continuity Agreement + Uncertainty effects = 0 over Feb-Dec 2019</i>						
Simple average			0.060 (0.430)	-0.995*** (0.222)	-1.080** (0.510)	-0.976*** (0.225)
Weighted average					0.252 (0.439)	-0.914*** (0.227)
Observations	550,435	550,435	550,435	550,435	549,992	549,992

Standard errors clustered at exporter-product level in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Columns 1, 3 and 5 report the results for the HL model, while columns 2, 4 and 6 report results for the PRF model. For the HL model, the tariff is the preferential tariff, while for the PRF model is the weighted average of the MFN and PRF rates. The dependent variable is the log of UK imports from PTA partners excluding processing trade. Uncertainty coefficients are estimated for each period between Jan2016-Dec2019, and year averages are reported. The Continuity Agreements effects are estimated for each month in columns 3-4 and for each cohort-month in columns 5-6, and averages are reported. For the HL model, uncertainty is measured as $1 - (\tau^{MFN}/\tau^{PRF})^\sigma$, with $\sigma = 4$. For the PRF model, this measure is multiplied by the share of preferential imports. The last panel reports the test of whether the sum of the average CA and the average uncertainty effects in Feb2019-Dec2019 equal zero. All models include partner-product-calendar month, partner-time and product-time fixed effects.

Figure 4: Uncertainty effects over time, PRF model



The figures report the coefficients and 95% confidence interval of the period-specific uncertainty effects of the PRF model based on the estimates reported in column 6 of Table 3.

2016. From then on uncertainty effects increase with some variation over time, with another large drop in December 2019, when the UK held its General Elections.

The increase in the (absolute) magnitude of uncertainty coefficients over time can be explained by sluggish response of the extensive margin to negative shocks. It is also possible that the probability of policy change increases over time. Considering the evolution of Brexit negotiations, which did not follow a straight path, it is likely that both forces are at play. A detailed discussion of the events that might have led to more uncertainty in 2019 is presented in section 6.2.

In Table 4 I use external information on the evolution of uncertainty over time to corroborate the results. In particular, I estimate a specification similar to Graziano et al. (2021) by interacting the product-level risk measure with the log of the Brexit Uncertainty Index developed by Bloom et al. (2017). I include up to two lags of the uncertainty index and report the cumulative effect. Details of this specification are reported in the Appendix.

The results are similar to the baseline ones. The HL model does not detect significant uncertainty effects, but it picks up a positive effects of the Continuity Agreements. On the other hand, the PRF model finds both evidence of uncertainty and Continuity Agreements effects. As for the baseline model, the Continuity Agreements effects are smaller than the uncertainty ones, suggesting that the CA removed only 26% of the uncertainty introduced by the 2016 referendum.

Table 4: Interaction with Brexit Uncertainty Index

	Model 1		Model 2	
	(1)	(2)	(3)	(4)
	HL	PRF	HL	PRF
MFN risk $\times \ln BUI_t$	0.335 (0.773)	-2.105*** (0.387)	0.170 (0.778)	-2.220*** (0.392)
CA $\times$ MFN risk $\times \ln BUI_t$			0.771*** (0.263)	0.587** (0.268)
Tariffs, lags 0-3	Yes	Yes	Yes	Yes
CA+MFN risk			0.942 (0.800)	-1.633*** (0.438)
Observations	549,992	549,992	549,992	549,992

Standard errors clustered at partner-CN8 level in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. In columns 1-2 the tariff is the preferential tariff, while in columns 3-4 it is the weighted-average of MFN and PRF rates. The dependent variable is the log of UK imports from PTA partners excluding processing trade. For the HL model, uncertainty is measured with the MFN risk $1 - (\tau^{MFN}/\tau^{PRF})^4$. For the PRF model the uncertainty measure is the one of the HL model multiplied by the PRF share. BUI is the Brexit Uncertainty Index from the Bank of England's Decision Maker Panel dataset. To deal with zeros, I take the hyperbolic sine transformation $\ln[x + (1 + x^2)^{0.5}]$. All models include the interaction of the MFN risk and the BUI up to two lags, and the sum of the interactions is reported. For Continuity Agreements, the MFN risk and BUI are interacted with a dummy for CA signature. The row "CA+MFN risk" reports the sum of the cumulative MFN and CA effects. All models include partner-product-calendar month, partner-time and product-time fixed effects.

6.1 Robustness tests

To corroborate the validity of the empirical results, I run a series of robustness tests. First, I consider the possible endogeneity of the share of preferential imports and of the potential profit loss ω_{hi} , which I address with instrumental variable regression. If Brexit uncertainty already started to affect the behaviour of exporters before the referendum, the baseline PRF share s_{hi}^{PRF} computed with data for 2013-15 might not reflect the true exposure to uncertainty and it might be correlated with the error term. I address this issue by instrumenting the PRF import share of the UK with the one of the EU27 $s_{hi,EU27}^{PRF}$. Because the UK and the EU are part of the same trade agreements, and since EU27's trade was not subject to Brexit uncertainty, the EU's PRF shares can serve as an instrument for UK's shares. The correlation between the UK's and EU27's PRF import share across partners and CN8 products is 0.70 in the period 2013-15. To maximise coverage, I use EU27 data for the period 2013-19. To further extend the coverage, I also compute the PRF share for the UK over the entire period 2013-19, and instrument it with the average 2013-19 EU27's share.

A second potential identification threat concerns the PRF/MFN ratio in ω , as recognised by Graziano et al. (2021). After leaving the EU, the UK would have set its own MFN tariff schedule. It can be argued that, as the UK was part of the EU and contributed to determining the EU MFN tariffs, a future UK MFN tariff schedule would have been very similar to the EU's one. However, there might have been some uncertainty around the new UK MFN tariff schedule, and ω_{hi} might have measurement error. I follow the approach of Graziano et al. (2021) and instrument ω_{hi} with the median of tail risks computed using the MFN tariffs of Australia, Canada, Japan and the US, which I call ω_{hi}^{IV} . If the UK PRF tariff was higher than the MFN tariff of the other countries, we would have the issue of $\omega_{hi}^{IV} < 0$, which is inconsistent with the theory. Hence, I replace $\omega_{hi}^{IV} = 0$ when it is negative. Note that to have a consistent product classification across countries we must rely on the HS 6-digit product classification rather than CN8. This means that variation in ω_{hi}^{IV} is at the 6-digit rather than 8-digit level, although regressions are done at the 8-digit level.

The results of the IV estimation, reported in Table 7 in the Appendix, confirm the baseline results. The effect of uncertainty remains negative and significant in all specifications, and increasing over time. Continuity Agreements effects are slightly larger than in the baseline, suggesting that the CAs removed 50% of the Brexit uncertainty.

To check if there are visible pre-treatment trends that can affect the main analysis, I run an event-study regression. The results are reported in Figure 7, and do not show particular signs of pre-trends.

In Table 13 I report additional results. Column 1 restricts the sample to countries which did not change the type of PTA with the EU over the sample period (labelled pre-2013 PTA). This could happen

if GSP countries moved to a Free Trade Agreement, or in cases such as South Africa changing its trade agreement with the EU in 2016.

In column 2 I control for shocks at the exporter-product level by using as dependent variable the difference between the log of UK imports and the log of EU27 imports. If there were shocks at the exporter-product level that drive the results but are not correlated with uncertainty, they should affect UK and EU imports in the same way. The results remain similar to the baseline.

Looking at product-level trade over multiple years, requires using concordance tables for product codes. This implies aggregating products codes that merge and split over time. To check if such aggregation creates any issues, in column 3 of Table 13 I exclude the products that merge and split over time.

The baseline specification excludes processing trade, as this is not subject to tariffs. In column 4 of Table 13 I include processing trade. Processing trade accounts only for a small fraction of total trade, and the empirical results are virtually unchanged compared to the baseline.

In Tables 14 and 15 I estimate the model with different levels of product-time fixed effects, both for the HL model (Table 14) and the PRF model (Table 15). Here I estimate uncertainty effects for each year between 2016-19 rather than each year-month, and the effect of Continuity Agreements is estimated with a single dummy for the CA interacted with the uncertainty measure. Using the HL model I find negative and significant uncertainty effects only without any product-time FE. When I include HS2-time FE, only the uncertainty effect in 2019 remains negative and significant, and with product-time FE at more disaggregated levels, all the uncertainty effects vanish. On the other hand, estimates of the CA effects remain stable and significant. For the PRF model (Table 15) the uncertainty effects are negative and significant across all specifications and tend to get bigger with finer levels of product-time FE. The CA effects remain stable across specifications. As discussed in section 6, the variation of preferential tariffs across countries is much more limited than the variation across products, while the share of preferential imports has similar variation both across and within products. In any case, product-time fixed effects can be important to absorb shocks affecting products over time common to all exporters.

Graziano et al. (2021) provide the only other estimate of Brexit uncertainty effects for PTA partners of the UK. For the sake of comparison, I estimate the specification with the Brexit Uncertainty Index – the most similar specification to GHL – on the same sample of PTA countries as in GHL: Iceland, Norway, Switzerland, Turkey, Mexico, Israel, Chile and South Korea. I report results based on both the HL and PRF models in Tables 16 and 17, respectively. As GHL does not include product-time fixed effects, I run regressions both without and with product-time fixed effects at different levels of product aggregation. Ignoring the share of PRF imports, I find a negative and significant effect of uncertainty only without product-time FE (see column 1 of Table 16), and the coefficient is similar to the -1.46 obtained by GHL. On the other hand, using the PRF model the uncertainty effects are negative and significant with and

without product-time FE, and increase in magnitude with product-time FE at more disaggregated product levels.

6.2 Interpretation and partial equilibrium quantification

This section presents an interpretation of results in light of the underlying theoretical model and a partial equilibrium quantification of the effects of uncertainty on UK imports from PTA partners.

The regression coefficients presented in this study provide a semi-elasticity measure of the effect of uncertainty on imports. The quantification exercise leverages these semi-elasticities, together with the theoretical model, to estimate the reduction in import values attributable to uncertainty. Given that uncertainty shocks impact import values primarily through the extensive margin – that is, the decision to trade under the Preferential or MFN regime – these effects take time to materialise fully in observed import values.

Although the referendum result in June 2016 represents the most significant uncertainty shock, other events have likely contributed to uncertainty fluctuations, leading to temporal variation in its intensity. Thus, the analysis incorporates two temporally varying components: (i) the lagged response of import values to an uncertainty shock due to gradual adjustments in the extensive margin, and (ii) the temporal variation in the intensity of uncertainty shocks.

By linking the estimated regression coefficients to model parameters, we can trace the evolution of uncertainty shocks over time. Second, the theoretical model allows us to compute the change in import values resulting from extensive margin responses to uncertainty. By combining these two elements, we can retrieve the response of import values to uncertainty over time.

The structural interpretation of the uncertainty coefficients in time T is $\tilde{b}_T^\gamma = -\frac{\alpha-\sigma+1}{\sigma-1}\psi\frac{\beta}{1-\beta}(1-\lambda)\left[\sum_{l=0}^{\infty}(1-\psi)^l\gamma_{T-l}\right]$, where α is the shape parameter of the firm distribution, σ is the elasticity of substitution, β is a discount factor, ψ is the (exogenous) probability of firm's exit, $(1-\lambda)$ is the probability of a bad draw for future tariffs and γ_T is the probability of policy change. The $\tilde{\cdot}$ refers to estimated coefficients. The uncertainty coefficient measures both current and lagged levels of γ . For the first uncertain period ($T = 1$) the only shock that occurred is γ_1 , while for $T > 1$ we can write the ratio of the uncertainty coefficient in period T and the one in period 1 as $\frac{\tilde{b}_T^\gamma}{\tilde{b}_1^\gamma} = \frac{\gamma_T}{\gamma_1} + \frac{\tilde{b}_{T-1}^\gamma}{\tilde{b}_1^\gamma}(1-\psi)$. Hence, with an assumption about the exit rate parameter ψ , we have an estimate of the probability of policy change over time relative to the first uncertain period:

$$\frac{\gamma_T}{\gamma_1} = \frac{\tilde{b}_T^\gamma}{\tilde{b}_1^\gamma} - \frac{\tilde{b}_{T-1}^\gamma}{\tilde{b}_1^\gamma}(1-\psi) \quad (14)$$

This allows us to see whether the pattern in the uncertainty coefficients seen in Figure 4 is mainly driven

by changes in the level of uncertainty over time or whether it is driven by the presence of legacy firms in the data.

As the monthly variation in the uncertainty and CA coefficients can contain quite some noise, I take the quarterly average coefficient and smooth the series with a LOWESS local smoother to focus on the trend of uncertainty over time. The smoothing result for the quarterly uncertainty coefficients is presented in Figure 9 in the Appendix. I then compute the evolution of the probability of policy γ_T over time relative to 2016q3 as per equation (14) based on the smoothed series. This is shown in panel a of Figure 5, together with the Continuity Agreements effect.¹⁴

For the partial equilibrium quantification, consider total trade in period t X_t as the sum of trade of new firms X_t^{new} and of the surviving legacy firms X_t^{sur} . Taking the ratio of trade under uncertain to the steady state equilibrium $\hat{X} = X_U/X_{ss}$ we have $\hat{X} = \theta_{ss}\hat{X}^{new} + (1 - \theta_{ss})\hat{X}^{sur}$, where θ_{ss} is the exports share of new firms at the steady state and $\hat{x} = x'/x$. At the steady state, past experience no longer matters and the share of surviving firms is given by the survival rate $(1 - \psi)$. Second, because uncertainty only affects the exporting decision of new firms, we have $\hat{X}^{sur} = 1$. Hence the counterfactual change in trade can be written as $\hat{X} = \psi\hat{X}^{new} + (1 - \psi)$. This is true for the first period in which uncertainty arises. If we consider the counterfactual T periods ahead from the steady state we have:

$$\hat{X}_T = \psi \sum_{l=0}^T (1 - \psi)^l \hat{X}_{T-l}^{new} + (1 - \psi)^{T+1} \quad (15)$$

The proportional change in the value of exports of new firms due to uncertainty with tariffs evaluated at the base period value is given by:

$$\hat{X}_T^{new} = \frac{1 + \left[\left(\tau_{hj}^{MFN} / \tau_{hj}^{PRF} \right)^\sigma - 1 \right] (F_{hj} \times U_T)^{-\alpha+\sigma-1}}{1 + \left[\left(\tau_{hj}^{MFN} / \tau_{hj}^{PRF} \right)^\sigma - 1 \right] (F_{hj})^{-\alpha+\sigma-1}} \quad (16)$$

where $U_T = \left(\frac{1-\beta+\beta\gamma_T}{1-\beta+\beta\gamma_T\lambda} \right)^{\frac{1}{\sigma-1}}$ for the MFN threat analysed in the empirical section. Tariff data are observed, and as for the empirical analysis I maintain the assumption $\sigma = 4$. In the Appendix I show that in equilibrium we have $F_{hj}^{-\alpha+\sigma-1} = \frac{X_{hj}^{PRF}}{X_{hj}^{PRF} + \left(\tau_{hj}^{MFN} / \tau_{hj}^{PRF} \right)^\sigma X_{hj}^{MFN}}$, which is a modified share or preferential trade adjusted for the tariff margin. Equation (14) shows how to derive γ_T for each period relative to the first period of uncertainty γ_1 . Hence, given a value of γ_1 we can derive the full path of γ . I follow Graziano et al. (2021) in assuming that the discount factor β is composed only of the exogenous probability of death

¹⁴I take the monthly average CA effects across cohorts and add them to the uncertainty coefficients. I then run the LOWESS smoothing on the series. For periods before 2019, I use the smoothed series on uncertainty only.

such that $\beta = 1 - \psi$. Then I can derive:

$$\gamma_1 (1 - \lambda) = \frac{-\tilde{b}_1^\gamma}{1 - \psi} \frac{\sigma - 1}{\alpha - \sigma + 1} \quad (17)$$

where $\tilde{b}_1^\gamma$ is the uncertainty coefficient for the first uncertainty period.

Calibration: To evaluate the term U_T in (16) I have to separate the probability of policy change γ from the probability of a bad draw from the tariff distribution $(1 - \lambda)$. I calibrate $1 - \lambda$ as the probability of a tariff being above the average preferential tariff. For each CN 8-digit product, I compute the average preferential tariff for the period 2013-15. I then construct a dataset that, for every product, contains the MFN and all preferential tariff rates, and compute the share of observations with a tariff below the average preferential for each CN8 product. This is done for products with non-zero MFN tariff. I then take the average of such share across all products, and I obtain a probability of being below the average preferential tariff of 0.17. I use this statistic to calibrate $\lambda = 0.17$.¹⁵

The parameter ψ measures the exit rate from the export market. Estimates of this parameter from firm-level studies vary depending on the unit of observation (e.g., firm or firm-destination). Alessandria et al. (2021) report survival rates after one year of about 80%, while Alborno et al. (2012) find that continuing firms (that exported in both year t and $t - 1$) accounted for 65-72% of exporters using Colombian data. I therefore set $\psi = 0.3$ at yearly frequency, which becomes 0.075 at quarterly frequency.

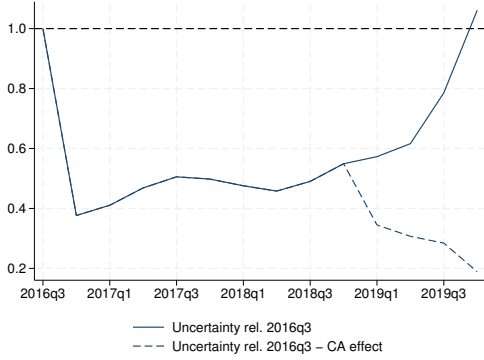
I maintain the assumption of $\sigma = 4$ used in the empirical analysis. For the shape parameter of the productivity distribution α , I follow Costinot and Rodríguez-Clare (2014) to assume $\alpha = 1.65 \times (\sigma - 1)$.

According to Figure 5, the major source of uncertainty has been the 2016 referendum. After that, the probability of policy change reduced sharply to around 40% of the 2016q3 level until mid-2018. From 2018q2 uncertainty rose again and slightly overpassed the referendum levels at the end of 2019.

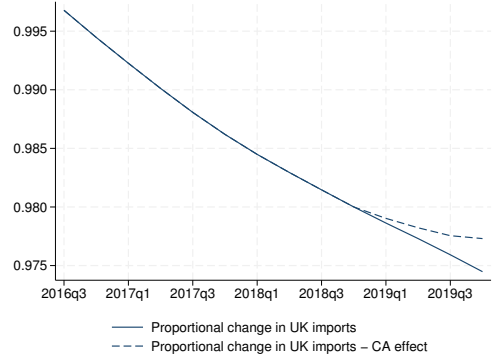
The estimated increase in uncertainty from mid-2018 is in line with the approaching of the Brexit deadline, originally set at the beginning of 2019. Moreover, the further increase in panel a of Figure 5 at the end of 2019 coincides with a series of events which most likely increased uncertainty regarding the UK-EU relation. Boris Johnson replaced Theresa May as prime minister in July 2019 when the Article 50 deadline for the UK exit was set on October 31. Johnson maintained a hard line on Brexit. He prorogued Parliament until October 14, although prorogation was then reversed, limiting its ability to discuss Brexit issues and he re-wrote part of the deal reached with the EU by May's government. The Brexit deadline

¹⁵I consider an 8-digit tariff line as the MFN tariff for an 8-digit product or the preferential tariff for an FTA-product to deal with some FTAs covering multiple countries.

Figure 5: Counterfactual results



(a) Probability of policy change relative to 2016q3



(b) Proportional change in UK imports

The figures report the estimated evolution of the probability of policy change over time relative to 2016q3 (panel a) and the proportional change in UK imports implied by the counterfactual based on the model's equations.

for a no-deal exit was moved to January 2020 only two days before the October 31 deadline. The country then held a general election on December 12. This series of events most likely increased uncertainty and the possibility of a policy reversal. At the same time, we see that the Continuity Agreements helped in keeping down the level of uncertainty at the moment when this was rising faster.

I compute the counterfactual loss in UK imports for each quarter between 2016q3-2019q4 based on equation (15). This is done at the sample average values of tariffs and PRF share as reported in Table 1. By the end of 2019, accounting for the cumulation of uncertainty over time, the reduction in imports relative to the no-uncertainty counterfactual is -2.6%. Adding in Continuity Agreements this effect is reduced to -2.3% in 2019q4.

If we consider a scenario in which the uncertainty of 2019q4 was there to stay, the long-run reduction in trade would have been -4.3%. In contrast, the HL model with firms only trading under the preferential regime would predict a change in imports of -7% instead of -4.3% (details in the Appendix). This shows the importance of taking into account the option of trading under MFN when evaluating the effects of uncertainty within PTAs.

7 Conclusion

This paper developed a model to study trade policy uncertainty within trade agreements. In PTAs, firms can choose between the standard MFN regime or, by paying an additional cost, access the preferential tariff regime.

The model predicts that uncertainty about future tariff preferences deters firms from using them. More firms decide to trade under the MFN regime, considering the option of switching to the preferential regime once uncertainty is resolved. As more firms pay the MFN tariff, the total trade value is reduced under uncertainty, and such reduction is proportional to the product of the margin between the MFN and preferential tariff and the preference utilisation rate.

The model predictions are validated through an empirical application to Brexit and UK imports from its PTA partners. A Wald test for nested models rejects the restriction of full preference utilisation imposed by the standard model, showing the importance of accounting for the information contained in preference utilisation rates when studying trade agreements.

The changes in trade due to uncertainty predicted by the model with both MFN and preferential trade are smaller than those of a model considering a single tariff regime. In the latter, uncertainty leads zero exports for some firms that take a wait-and-see approach, while in my model firms keep exporting but under the higher MFN tariff rate. This makes the changes in trade smaller, with consequences for the welfare effects of trade policy uncertainty.

The empirical analysis indicates a decline of -2.6% in UK imports from PTA partners by the end of 2019. Continuity Agreements signed to replace the existing EU trade agreements removed 30-50% of the uncertainty generated by the referendum. My results show that when studying uncertainty about conditional tariff preferences, incorporating preference utilisation data can lead to substantially different empirical results. In general, when firms serve a market under different regimes, regime heterogeneity should be taken into account.

Beyond the case of Brexit, the model could be applied to study the uncertainty generated by the conditionality of GSP schemes. It could be applied to the NAFTA renegotiation started in 2017-18, but also to evaluate the new ‘sunset’ clause of the USMCA (the new NAFTA) for which the agreement must be renegotiated periodically. In a time of increased geopolitical turmoil, this paper offers a model to evaluate the effects of policy uncertainty in trade agreements, expanding the suite of models for applied work on trade policy uncertainty.

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Appendix for online publication

A Theory

Model setting

Consumers aggregate consumption between a freely traded homogeneous good q_0 and a differentiated good Q with Cobb-Douglas preferences $U = q_0^{1-\mu} Q^\mu$. The aggregation of different varieties q_h is done with CES utility function $Q = [\int q_h^\rho dh]^{1/\rho}$ characterised by elasticity of substitution $\sigma = 1/(1 - \rho) > 1$. Solving the utility maximisation problem gives the demand for a variety h from country i in country j as $q_{hijt} = p_{hijt}^{-\sigma} \mu Y_{jt} P_{jt}^{\sigma-1}$ where Y_{jt} is aggregate income of country j at time t , $P_{jt} = [\int p_{hjt}^{1-\sigma} dh]^{1/(1-\sigma)}$ is the CES price index and the price p_{hijt} is inclusive of trade costs $\delta_{ij} \geq 1$ and tariffs $\tau_{hijt} \geq 1$.

Firms are monopolistic competitors using labour as the only mean of production, which cost is pinned down to unity by the numeraire. They are also differentiated by their productivity level φ , and they maximise the profit $q_{hijt} (p_{hijt} - \delta_{ij} \tau_{hijt} / \varphi)$ yielding the optimal price as a constant markup over variable costs $p_{hijt} = \frac{\sigma}{\sigma-1} \frac{\delta_{ij} \tau_{hijt}}{\varphi}$, which depends on the firm's productivity φ . Using the residual demand and the price received by the firm (i.e., excluding tariffs) we can determine the operating profit of a firm as:

$$\pi_{hijt} = q_{hijt} p_{hijt} / \tau_{hijt} = \left(\frac{\sigma}{\sigma-1} \frac{\delta_{ij}}{\varphi} \right)^{1-\sigma} \tau_{hijt}^{-\sigma} \mu Y_{jt} P_{jt}^{\sigma-1} \quad (18)$$

Note that in the profit function (18) the tariff enters with an exponent of $-\sigma$, differently from the one on the iceberg transport cost δ_{ij} which is $1 - \sigma$. This is because while transport costs can be passed to consumers, tariff revenues are collected by the customs authorities. Hence, the firm charging a price $p_{ij} = p_i \tau_{ij}$ to consumers will only receive a value p_i for each unit sold.

To start exporting from i to j a firm incurs an initial sunk cost. If countries i and j do not have a PTA the only option available to an exporter is the MFN regime. In this case the firm would pay sunk costs f_{hij} and face the MFN tariff τ_{hjt}^{MFN} . If on the other hand countries i and j have a PTA in place the firm can choose to export under the MFN or the preferential (PRF) regime. If she chooses to use preferences there is an additional sunk cost f_{hij}^{PRF} . These additional costs cover the research costs to understand which documents are necessary to use preferences, comply with bureaucracy etc. Hence sunk costs under the PRF regime are $f_{hij} + f_{hij}^{PRF}$. Under the PRF regime the firm has an advantage in variable costs as she faces the preferential tariff $\tau_{hijt}^{PRF} \leq \tau_{hjt}^{MFN}$. Upon entry, the firm decides whether to enter and under which strategy by maximising the present value of exporting. Given a discount factor $\beta < 1$, in the deterministic scenario the present values of the MFN and PRF regimes are:

$$V_{hijt}^{MFN} = \pi(\tau_{hjt}^{MFN}) + \sum_{t=1}^{\infty} \beta^t \pi(\tau_{hjt}^{MFN}) = \frac{\pi(\tau_{hjt}^{MFN})}{1 - \beta}$$

$$V_{hijt}^{PRF} = \pi(\tau_{hijt}^{PRF}) + \sum_{t=1}^{\infty} \beta^t \pi(\tau_{hijt}^{PRF}) = \frac{\pi(\tau_{hijt}^{PRF})}{1 - \beta}$$

The firm exports under the MFN regime if $V_{hijt}^{MFN} \geq f_{hij}$, while she chooses the PRF regime if $V_{hijt}^{PRF} - V_{hijt}^{MFN} \geq f_{hij} + f_{hij}^{PRF} - f_{hij} = f_{hij}^{PRF}$. The choice between the two regimes depends on the productivity of the firm. Because the PRF regime involves higher fixed costs but lower variable costs than the MFN regime, only the most productive firms will chose the PRF regime. This is because most productive firms will have higher sales volumes, so they can distribute the higher sunk costs across more units of product sold. This is the same result of Helpman et al. (2004) in the export/FDI decision. We can determine the productivity cut-off above which the firm exports under the MFN regime φ_{hijt}^{MFN} by solving $V_{hijt}^{MFN} = f_{hij}$ for productivity:

$$\varphi_{hijt}^{MFN} = \left[\frac{f_{hij} (1 - \beta)}{A_{ijt} (\tau_{hjt}^{MFN})^{-\sigma}} \right]^{\frac{1}{\sigma-1}}$$

where $A_{ijt} = \left(\frac{\sigma}{\sigma-1} \delta_{ij} \right)^{1-\sigma} \mu Y_{jt} P_{jt}^{\sigma-1}$ summarises market conditions. The cut-off for the PRF regime φ_{hijt}^{PRF} can be found solving $V_{hijt}^{PRF} - V_{hijt}^{MFN} = f_{hij}^{PRF}$ for φ :

$$\varphi = \left[\frac{f_{hij}^{PRF} (1 - \beta)}{A_{ijt} \left[(\tau_{hjt}^{PRF})^{-\sigma} - (\tau_{hjt}^{MFN})^{-\sigma} \right]} \right]^{\frac{1}{\sigma-1}}$$

Then collect $(\tau_{hjt}^{MFN})^{-\sigma}$ at the denominator and divide and multiply the numerator by f_{hij} to have:

$$\varphi_{hijt}^{PRF} = \left[\frac{f_{hij} (1 - \beta)}{A_{ijt} (\tau_{hjt}^{MFN})^{-\sigma}} \right]^{\frac{1}{\sigma-1}} \left[\frac{f_{hij}^{PRF} / f_{hij}}{(\tau_{hjt}^{MFN} / \tau_{hijt}^{PRF})^{\sigma} - 1} \right]^{\frac{1}{\sigma-1}}$$

where the second term measures the ratio of the proportional increase in fixed costs f_{hij}^{PRF} / f_{hij} over the proportional saving in variable costs $(\tau_{hjt}^{MFN} / \tau_{hijt}^{PRF})^{\sigma} - 1$. It is clear that we have $\varphi_{hijt}^{PRF} > \varphi_{hijt}^{MFN}$, and therefore observe MFN trade, if and only if

$$f_{hij}^{PRF} / f_{hij} > (\tau_{hjt}^{MFN} / \tau_{hjt}^{PRF})^\sigma - 1$$

This condition is very general and it states that firms will choose the MFN regime only if the proportional increase in fixed costs associated with the PRF regime is larger than the proportional increase in prices given by the MFN regime. Note that for the empirical application, if no firm trade under the MFN regime then we are back to the HL model with only one trade regime.

Cutoff under uncertainty

To find the cutoff of the PRF regime under uncertainty, start solving recursively equation (3):

$$V_{hijt}^{PRF} = \frac{\pi(\tau_{hijt}^{PRF})}{1 - \beta + \beta\gamma} + \frac{\beta\gamma}{1 - \beta} \frac{E_t\pi(\tau')}{1 - \beta + \beta\gamma} \quad (19)$$

where we made use of $E_t V(\tau') = E_t \pi(\tau') / (1 - \beta)$.

For the value of the MFN strategy in equation (4), note that:

$$\begin{aligned} E_t V(\tau' | \tau' \leq \bar{\tau}) &= E_t \pi(\tau' | \tau' \leq \bar{\tau}) + \beta[(1 - \gamma)E_t V(\tau' | \tau' \leq \bar{\tau}) + \gamma E_t V(\tau')] = \\ &= \frac{E_t \pi(\tau' | \tau' \leq \bar{\tau})}{1 - \beta + \beta\gamma} + \frac{\beta\gamma}{1 - \beta} \frac{E_t \pi(\tau')}{1 - \beta + \beta\gamma} \end{aligned}$$

Substituting this expression into (4) and solving for V_{hijt}^{MFN} we obtain:

$$V_{hijt}^{MFN} = \frac{\pi(\tau_{hijt}^{MFN})}{1 - \beta + \beta\gamma H(\bar{\tau})} + \frac{\beta\gamma H(\bar{\tau})}{1 - \beta + \beta\gamma H(\bar{\tau})} \left[\frac{E_t \pi(\tau' | \tau' \leq \bar{\tau})}{1 - \beta + \beta\gamma} + \frac{\beta\gamma}{1 - \beta} \frac{E_t \pi(\tau')}{1 - \beta + \beta\gamma} - f_{hij}^{PRF} \right] \quad (20)$$

Next, set the difference between (19) and (20) equal to f_{hij}^{PRF} and rearrange to get:

$$\begin{aligned} &\frac{\pi(\tau_{hijt}^{PRF})(1 - \beta)(1 - \beta + \beta\gamma H(\bar{\tau})) + \beta\gamma E_t \pi(\tau')(1 - \beta + \beta\gamma H(\bar{\tau}))}{(1 - \beta)(1 - \beta + \beta\gamma)(1 - \beta + \beta\gamma H(\bar{\tau}))} + \\ &\quad - \frac{\pi(\tau_{hijt}^{MFN})(1 - \beta)(1 - \beta + \beta\gamma) + \beta\gamma H(\bar{\tau}) E_t \pi(\tau' | \tau' \leq \bar{\tau})(1 - \beta)}{(1 - \beta)(1 - \beta + \beta\gamma)(1 - \beta + \beta\gamma H(\bar{\tau}))} + \\ &\quad - \frac{\beta\gamma H(\bar{\tau}) \beta\gamma E_t \pi(\tau')}{(1 - \beta)(1 - \beta + \beta\gamma)(1 - \beta + \beta\gamma H(\bar{\tau}))} = f_{hij}^{PRF} \left(1 - \frac{\beta\gamma H(\bar{\tau})}{1 - \beta + \beta\gamma H(\bar{\tau})} \right) \end{aligned}$$

Simplifying we obtain

$$\frac{[\pi(\tau_{hijt}^{PRF}) - \pi(\tau_{hijt}^{MFN})](1 - \beta)}{1 - \beta + \beta\gamma} + \beta\gamma \frac{H(\bar{\tau})\pi(\tau_{hijt}^{PRF}) - \pi(\tau_{hijt}^{MFN}) + E_t\pi(\tau') - H(\bar{\tau})E_t\pi(\tau'|\tau' \leq \bar{\tau})}{1 - \beta + \beta\gamma} = f_{hij}^{PRF}(1 - \beta)$$

Note that $E_t\pi(\tau') - H(\bar{\tau})E_t\pi(\tau'|\tau' \leq \bar{\tau}) = (1 - H(\bar{\tau}))E_t\pi(\tau'|\tau' > \bar{\tau})$. Moreover, recalling that profits can be expressed as $\pi = A\tau^{-\sigma}\varphi^{\sigma-1}$, solving for productivity φ we obtain:

$$\varphi^{\sigma-1} = \frac{f_{hij}^{PRF}(1 - \beta)}{A_{ijt}} \frac{1 - \beta + \beta\gamma}{\left[(\tau_{hijt}^{PRF})^{-\sigma} - (\tau_{hijt}^{MFN})^{-\sigma}\right](1 - \beta) + \beta\gamma \left[H(\bar{\tau})(\tau_{hijt}^{PRF})^{-\sigma} - (\tau_{hijt}^{MFN})^{-\sigma} + (1 - H(\bar{\tau}))E_t(\tau'|\tau' > \bar{\tau})\right]}$$

Collecting $(\tau_{hijt}^{PRF})^{-\sigma} - (\tau_{hijt}^{MFN})^{-\sigma}$ at the denominator of the second fraction, and rearranging, we get:

$$\varphi^{\sigma-1} = \frac{f_{hij}^{PRF}(1 - \beta)}{A_{ijt}} \frac{1}{(\tau_{hijt}^{PRF})^{-\sigma} - (\tau_{hijt}^{MFN})^{-\sigma}} \frac{1 - \beta + \beta\gamma}{1 - \beta + \beta\gamma \left[1 - (1 - H(\bar{\tau})) \frac{(\tau_{hijt}^{PRF})^{-\sigma} - E_t(\tau'|\tau' > \bar{\tau})}{(\tau_{hijt}^{PRF})^{-\sigma} - (\tau_{hijt}^{MFN})^{-\sigma}}\right]}$$

Finally, rearranging and substituting the definition of ω_{hijt}^{PRF} we have:

$$\varphi^{UPRF} = \left[\frac{f_{hij}(1 - \beta)}{A_{ijt}(\tau_{hijt}^{MFN})^{-\sigma}} \right]^{\frac{1}{\sigma-1}} \left[\frac{f_{hij}^{PRF}/f_{hij}}{(\tau_{hijt}^{MFN}/\tau_{hijt}^{PRF})^{\sigma} - 1} \right]^{\frac{1}{\sigma-1}} \left[\frac{1 - \beta + \beta\gamma}{1 - \beta + \beta\gamma\omega_{hijt}^{PRF}} \right]^{\frac{1}{\sigma-1}}$$

Proof that $\omega_{hijt}^{PRF} < 1$

This section proves that $\omega_{hijt}^{PRF} < 1$. Starting from the definition of ω_{hijt}^{PRF} we have:

$$\begin{aligned} \omega_{hijt}^{PRF} &= 1 - (1 - H(\tau_{hijt}^{PRF})) \frac{(\tau_{hijt}^{PRF})^{-\sigma} - E_t(\tau^{-\sigma}|\tau' > \tau_{hijt}^{PRF})}{(\tau_{hijt}^{PRF})^{-\sigma} - (\tau_{hijt}^{MFN})^{-\sigma}} < 1 \\ &\quad - (1 - H(\tau_{hijt}^{PRF})) \frac{(\tau_{hijt}^{PRF})^{-\sigma} - E_t(\tau^{-\sigma}|\tau' > \tau_{hijt}^{PRF})}{(\tau_{hijt}^{PRF})^{-\sigma} - (\tau_{hijt}^{MFN})^{-\sigma}} < 0 \end{aligned}$$

Since $\tau_{hijt}^{PRF} \leq \tau_{hijt}^{MFN}$, $\tau_{hijt}^{PRF} < E_t(\tau^{-\sigma}|\tau' > \tau_{hijt}^{PRF})$ and $\sigma > 1$, both the nominator and the denominator in the fraction on the LHS are positive. And because $H(\tau_{hijt}^{PRF}) \in (0, 1)$, the LHS is negative. This completes the proof. In case only positive draws are possible ($H(\tau_{hijt}^{PRF}) = 1$) there is no uncertainty and in this case $\omega_{hijt}^{PRF} = 1$ and we are back to the deterministic case.

Derivation of exports value under uncertainty for new firms:

To find the value of exports of firms arriving in period t , integrate firm sales over firm productivity, which has a Pareto distribution $G(\varphi) = 1 - \varphi^{-\alpha}$. The value of PRF exports is then the value of exports of all firms above the PRF cut-off:

$$\begin{aligned} X_{hijt}^{PRF} &= A_{ijt} \left(\tau_{hijt}^{PRF} \right)^{-\sigma} m_{it} \int_{\varphi_{UPRF}}^{\infty} \varphi^{\sigma-1} dG(\varphi) = A_{ijt} \left(\tau_{hijt}^{PRF} \right)^{-\sigma} m_{it} \frac{\alpha}{\alpha-\sigma+1} \left(\varphi_{hijt}^{UPRF} \right)^{-\alpha+\sigma-1} \\ X_{hijt}^{PRF} &= A_{ijt} \left(\tau_{hijt}^{PRF} \right)^{-\sigma} m_{it} \frac{\alpha}{\alpha-\sigma+1} \left(\varphi_{hijt}^{MFN} \times F_{hijt} \times U_t \right)^{-\alpha+\sigma-1} \\ X_{hijt}^{PRF} &= A_{ijt}^{\frac{\alpha}{\sigma-1}} \left(\tau_{hjt}^{MFN} / \tau_{hijt}^{PRF} \right)^{\sigma} \left(\tau_{hjt}^{MFN} \right)^{-\frac{\sigma\alpha}{\sigma-1}} m_{it} \frac{\alpha}{\alpha-\sigma+1} [f_{hij} (1-\beta)]^{\frac{-\alpha+\sigma-1}{\sigma-1}} (F_{hijt} \times U_t)^{-\alpha+\sigma-1} \end{aligned}$$

For the value of exports under the MFN regime we have to integrate firms' revenues between the MFN and PRF cut-offs:

$$\begin{aligned} X_{hijt}^{MFN} &= A_{jt} \left(\tau_{hjt}^{MFN} \right)^{-\sigma} m_{it} \int_{\varphi_{MFN}}^{\varphi_{UPRF}} \varphi^{\sigma-1} dG(\varphi) \\ X_{hijt}^{MFN} &= A_{jt} \left(\tau_{hjt}^{MFN} \right)^{-\sigma} m_{it} \frac{\alpha}{\alpha-\sigma+1} \left[\left(\varphi_{hijt}^{MFN} \right)^{-\alpha+\sigma-1} - \left(\varphi_{hijt}^{UPRF} \right)^{-\alpha+\sigma-1} \right] \\ X_{hijt}^{MFN} &= A_{jt} \left(\tau_{hjt}^{MFN} \right)^{-\sigma} m_{it} \frac{\alpha}{\alpha-\sigma+1} \left(\varphi_{hijt}^{MFN} \right)^{-\alpha+\sigma-1} \left[1 - (F_{hijt} \times U_t)^{-\alpha+\sigma-1} \right] \\ X_{hijt}^{MFN} &= A_{ijt}^{\frac{\alpha}{\sigma-1}} \left(\tau_{hjt}^{MFN} \right)^{-\frac{\sigma\alpha}{\sigma-1}} m_{it} \frac{\alpha}{\alpha-\sigma+1} [f_{hij} (1-\beta)]^{\frac{-\alpha+\sigma-1}{\sigma-1}} \left[1 - (F_{hijt} \times U_t)^{-\alpha+\sigma-1} \right] \end{aligned}$$

We can then find total exports of new firms by adding X_{hijt}^{MFN} and X_{hijt}^{PRF} . After rearranging we have:

$$X_{hijt} = A_{ijt}^{\frac{\alpha}{\sigma-1}} \left(\tau_{hjt}^{MFN} \right)^{-\frac{\sigma\alpha}{\sigma-1}} m_{it} \frac{\alpha}{\alpha-\sigma+1} [f_{hij} (1-\beta)]^{\frac{-\alpha+\sigma-1}{\sigma-1}} \times \left[1 + \left(\left(\frac{\tau_{hjt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^{\sigma} - 1 \right) (F_{hijt} \times U_t)^{-\alpha+\sigma-1} \right] \quad (21)$$

Recalling that $A_{ijt} = \left(\frac{\sigma}{\sigma-1} \delta_{ij} \right)^{1-\sigma} \mu Y_{jt} P_{jt}^{\sigma-1}$ we can collect the following terms: $B_{jt} = Y_{jt} P_{jt}^{\sigma-1}$ summarises constant terms and aggregate demand conditions, m_{it} summarises supply conditions and

$$C_{hij} = \left(\frac{\alpha}{\alpha-\sigma+1} \left(\frac{\sigma}{\sigma-1} \right)^{1-\sigma} \mu \right)^{\frac{\alpha}{\sigma-1}} \delta_{ij}^{-\alpha} [f_{hij} (1-\beta)]^{\frac{-\alpha+\sigma-1}{\sigma-1}}$$

summarises time-invariant barriers to trade. Substituting the definitions of B_{jt} and C_{hij} in (21) yields equation (8) in the text.

Semi-elasticity with respect to γ

The semi-elasticity of trade of new firms with respect to the probability of a policy change is derived by computing the partial derivative of the log of (21) with respect to γ :

$$\frac{\partial \ln X_{hijt}}{\partial \gamma} = \frac{\partial \ln X_{hijt}}{\partial U} \frac{\partial U}{\partial \gamma} = -(\alpha - \sigma + 1) \frac{\left[\left(\frac{\tau_{hijt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^\sigma - 1 \right] (F_{hijt} \times U_t)^{-\alpha + \sigma - 1}}{1 + \left[\left(\frac{\tau_{hijt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^\sigma - 1 \right] (F_{hijt} \times U_t)^{-\alpha + \sigma - 1}} \frac{1}{U_t} \frac{\partial U}{\partial \gamma} \quad (22)$$

Recalling that $U = \left[\frac{1 - \beta + \beta \gamma}{1 - \beta + \beta \gamma \omega_{hijt}^{PRF}} \right]^{\frac{1}{\sigma - 1}}$ we have:

$$\begin{aligned} \frac{\partial U}{\partial \gamma} &= \frac{1}{\sigma - 1} \left[\frac{1 - \beta + \beta \gamma}{1 - \beta + \beta \gamma \omega_{hijt}^{PRF}} \right]^{\frac{1}{\sigma - 1} - 1} \frac{\beta (1 - \beta + \beta \gamma \omega_{hijt}^{PRF}) - \beta \omega_{hijt}^{PRF} (1 - \beta + \beta \gamma)}{(1 - \beta + \beta \gamma \omega_{hijt}^{PRF})^2} = \\ &= \frac{1}{\sigma - 1} U \frac{1 - \beta + \beta \gamma \omega_{hijt}^{PRF}}{1 - \beta + \beta \gamma} \beta \frac{(1 - \beta + \beta \gamma \omega_{hijt}^{PRF}) - \omega_{hijt}^{PRF} (1 - \beta + \beta \gamma)}{(1 - \beta + \beta \gamma \omega_{hijt}^{PRF})^2} = \\ &= \frac{1}{\sigma - 1} U \beta \frac{(1 - \beta) (1 - \omega_{hijt}^{PRF})}{(1 - \beta + \beta \gamma) (1 - \beta + \beta \gamma \omega_{hijt}^{PRF})} \end{aligned} \quad (23)$$

Where the second line is derived recognising that $U = \left[\frac{1 - \beta + \beta \gamma}{1 - \beta + \beta \gamma \omega_{hijt}^{PRF}} \right]^{\frac{1}{\sigma - 1}}$ and factoring β in the last fraction. The third line simplifies terms. Substituting (23) into (22) and multiplying by $\left(\frac{\tau_{hijt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^\sigma$ we have:

$$\begin{aligned} \frac{\partial \ln X_{hijt}}{\partial \gamma} &= - \left(\frac{\alpha - \sigma + 1}{\sigma - 1} \right) \frac{\left(\frac{\tau_{hijt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^\sigma (F_{hijt} \times U_t)^{-\alpha + \sigma - 1}}{1 + \left[\left(\frac{\tau_{hijt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^\sigma - 1 \right] (F_{hijt} \times U_t)^{-\alpha + \sigma - 1}} \\ &\quad \left[1 - \left(\frac{\tau_{hijt}^{PRF}}{\tau_{hijt}^{MFN}} \right)^\sigma \right] \beta \frac{(1 - \beta) (1 - \omega_{hijt}^{PRF})}{(1 - \beta + \beta \gamma) (1 - \beta + \beta \gamma \omega_{hijt}^{PRF})} \end{aligned} \quad (24)$$

Given the definition of ω_{hijt}^{PRF} of equation (6) we have:

$$1 - \omega_{hijt}^{PRF} = (1 - H(\tau_{hijt}^{PRF})) \frac{1 - E_t \left(\tau^{-\sigma} | \tau' > \tau_{hijt}^{PRF} \right) / \left(\tau_{hijt}^{PRF} \right)^{-\sigma}}{1 - \left(\tau_{hijt}^{PRF} / \tau_{hijt}^{MFN} \right)^\sigma}$$

Plugging this result into (24) and simplifying we obtain equation (9) in the text.

The value of exports with legacy firms

In product-level trade data at each point in time the observed value of exports is the sum of exports of new firms and exports of firms that entered in the past and are still trading today. Hence, we observe the following quantities for PRF and MFN trade in the data:

$$X_{hijt}^{PRF} = A_{ijt} \left(\tau_{hijt}^{PRF} \right)^{-\sigma} \frac{\alpha}{\alpha - \sigma + 1} \sum_{l=0}^{\infty} (1 - \psi)^l m_{it-l} \left(\varphi_{hijt-l}^{UPRF} \right)^{-\alpha + \sigma - 1} \quad (25)$$

$$X_{hijt}^{MFN} = A_{ijt} \left(\tau_{hijt}^{MFN} \right)^{-\sigma} \frac{\alpha}{\alpha - \sigma + 1} \sum_{l=0}^{\infty} (1 - \psi)^l m_{it-l} \left(\varphi_{hijt-l}^{MFN} \right)^{-\alpha + \sigma - 1} \left[1 - (F_{hijt-l} U_{t-l})^{-\alpha + \sigma - 1} \right] \quad (26)$$

where l stands for lags and ψ is the exogenous probability of death of a firm. Total trade is the sum of PRF and MFN trade:

$$X_{hijt} = A_{ijt} \frac{\alpha}{\alpha - \sigma + 1} \left(\tau_{hijt}^{MFN} \right)^{-\sigma} \times \sum_{l=0}^{\infty} (1 - \psi)^l m_{it-l} \left[\left(\frac{\tau_{hijt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^{\sigma} \left(\varphi_{hijt-l}^{UPRF} \right)^{-\alpha + \sigma - 1} + \left(\varphi_{hijt-l}^{MFN} \right)^{-\alpha + \sigma - 1} \left(1 - (F_{hijt-l} U_{t-l})^{-\alpha + \sigma - 1} \right) \right]$$

considering that $\varphi_{hijt-l}^{UPRF} = \varphi_{hijt-l}^{MFN} \times F_{hijt-l} \times U_{t-l}$ and regrouping terms within the sum:

$$X_{hijt} = A_{ijt} \frac{\alpha}{\alpha - \sigma + 1} \left(\tau_{hijt}^{MFN} \right)^{-\sigma} \times \left\{ \sum_{l=0}^{\infty} (1 - \psi)^l m_{it-l} \left(\varphi_{hijt-l}^{MFN} \right)^{-\alpha + \sigma - 1} \left[1 + \left(\left(\frac{\tau_{hijt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^{\sigma} - 1 \right) (F_{hijt-l} U_{t-l})^{-\alpha + \sigma - 1} \right] \right\}$$

substituting $\varphi_{hijt-l}^{MFN} = \left[\frac{f_{hij}(1-\beta)}{A_{ijt}(\tau_{hijt}^{MFN})^{-\sigma}} \right]^{\frac{1}{\sigma-1}}$ and using the definitions of B_{jt} and C_{hij} given above we have equation (10) in the text.

B Derivation of the empirical model

I take the log-linear approximation of equation (10) around $B_{jt-l} = B_{j0}$, $\tau_{hjt-l}^{MFN} = \tau_{hj0}^{MFN}$, $\tau_{hijt-l}^{PRF} = \tau_{hij0}^{PRF}$, $m_{it-l} = m_{i0}$ and $\gamma_{t-l} = 0$ for every l . The following paragraphs report the partial derivatives of the log of

(10) with respect to each time varying term, and then evaluate them at the equilibrium point 0. We have:

$$\begin{aligned} \ln X_{hijt|t=0, \gamma=0} &= \frac{\alpha}{\sigma-1} \ln B_{j0} + \ln C_{hij} + \ln m_{i0} - \frac{\sigma\alpha}{\sigma-1} \ln \tau_{hj0}^{MFN} \\ &\quad \ln \left[1 + \left(\left(\frac{\tau_{hj0}^{MFN}}{\tau_{hj0}^{PRF}} \right)^\sigma - 1 \right) (F_{hij0})^{-\alpha+\sigma-1} \right] + \ln \frac{1}{\psi} + \\ &\quad \sum_{l=0}^{\infty} \frac{\partial \ln X_{hijt}}{\partial \ln B_{jt-l}} \ln \left(\frac{B_{jt-l}}{B_{j0}} \right) + \sum_{l=0}^{\infty} \frac{\partial \ln X_{hijt}}{\partial \ln m_{it-l}} \ln \left(\frac{m_{it-l}}{m_{i0}} \right) + \\ &\quad \sum_{l=0}^{\infty} \frac{\partial \ln X_{hijt}}{\partial \ln \tau_{hjt-l}^{MFN}} \ln \left(\frac{\tau_{hjt-l}^{MFN}}{\tau_{hj0}^{MFN}} \right) + \sum_{l=0}^{\infty} \frac{\partial \ln X_{hijt}}{\partial \ln \tau_{hjt-l}^{PRF}} \ln \left(\frac{\tau_{hjt-l}^{PRF}}{\tau_{hj0}^{PRF}} \right) + \\ &\quad \sum_{l=0}^{\infty} \frac{\partial \ln X_{hijt}}{\partial \gamma_{t-l}} \gamma_{t-l} \end{aligned}$$

Evaluating the derivatives at the steady state with no uncertainty we obtain:

$$\begin{cases} \frac{\partial \ln X_{hijt}}{\partial \ln B_{jt-l}} = \frac{\alpha-\sigma+1}{\sigma-1} (1-\psi)^l \psi & \text{Importer conditions} \\ \frac{\partial \ln X_{hijt}}{\partial \ln m_{it-l}} = (1-\psi)^l \psi & \text{Exporter conditions} \\ \frac{\partial \ln X_{hijt}}{\partial \ln \tau_{hjt}^{MFN}} = -\frac{\alpha\sigma}{\sigma-1} \psi (1-s_{hij}^{PRF}) - \sigma (1-\psi) (1-s_{hij}^{PRF}) & \text{Current MFN tariff} \\ \frac{\partial \ln X_{hijt}}{\partial \ln \tau_{hjt-l}^{MFN}} = -\sigma \frac{\alpha-\sigma+1}{\sigma-1} \psi (1-\psi)^l (1-s_{hij}^{PRF}) & \text{Lagged MFN tariff} \\ \frac{\partial \ln X_{hijt}}{\partial \ln \tau_{hijt}^{PRF}} = -\frac{\alpha\sigma}{\sigma-1} \psi s_{hij}^{PRF} - \sigma (1-\psi) s_{hij}^{PRF} & \text{Current PRF tariff} \\ \frac{\partial \ln X_{hijt}}{\partial \ln \tau_{hijt-l}^{PRF}} = -\sigma \frac{\alpha-\sigma+1}{\sigma-1} \psi (1-\psi)^l s_{hij}^{PRF} & \text{Lagged PRF tariff} \\ \frac{\partial \ln X_{hijt}}{\partial \gamma_{t-l}} = -\frac{\alpha-\sigma+1}{\sigma-1} \psi (1-\psi)^l \frac{\beta}{1-\beta} (1-H(\tau_{hijt}^{PRF})) s_{hij}^{PRF} \left[1 - \left(\frac{\tau_{hj0}^{PRF}}{\tau_{hj0}^{MFN}} \right)^\sigma \right] & \text{Uncertainty} \end{cases}$$

For the uncertainty term, the derivation is as follows:

$$\begin{aligned} \frac{\partial \ln X_{hijt}}{\partial \gamma_t} &= -\frac{\alpha-\sigma+1}{\sigma-1} \times \\ &\quad \frac{m_{it} B_{jt}^{\frac{\alpha-\sigma+1}{\sigma-1}} \left(\tau_{hjt}^{MFN} \right)^{-\sigma \frac{\alpha-\sigma+1}{\sigma-1}} \left(\left(\frac{\tau_{hjt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^\sigma - 1 \right) \frac{-\alpha+\sigma-1}{\sigma-1} (F_{hijt} \times U_t)^{-\alpha+\sigma-1} U_t^{-1}}{\sum_{l=0}^{\infty} (1-\psi)^l m_{it-l} B_{jt-l}^{\frac{\alpha-\sigma+1}{\sigma-1}} \left(\tau_{hjt-l}^{MFN} \right)^{-\sigma \frac{\alpha-\sigma+1}{\sigma-1}} \left[1 + \left(\left(\frac{\tau_{hjt-l}^{MFN}}{\tau_{hijt}^{PRF}} \right)^\sigma - 1 \right) (F_{hijt-l} \times U_{t-l})^{-\alpha+\sigma-1} \right]} \times \\ &\quad \frac{\beta \left(1 - \beta + \beta \gamma_t \omega_{hij}^{PRF} \right) - \beta \omega_{hij} (1 - \beta + \beta \gamma_t)}{\left(1 - \beta + \beta \gamma_t \omega_{hij}^{PRF} \right)^2} \end{aligned}$$

Evaluating at the steady state and $\gamma = 0$ and taking out time-invariant terms from the sum at the denominator of the second line and simplifying:

$$\frac{\partial \ln X_{hijt}}{\partial \gamma_t} = -\frac{\alpha - \sigma + 1}{\sigma - 1} \left[\left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^\sigma - 1 \right] \frac{(F_{hij0})^{-\alpha + \sigma - 1}}{1 + \left[\left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^\sigma - 1 \right]} \psi \frac{\beta}{1 - \beta} (1 - \omega_{hij}^{PRF})$$

Rearrange the second and third terms to have:

$$\begin{aligned} \frac{\partial \ln X_{hijt}}{\partial \gamma_t} = & -\frac{\alpha - \sigma + 1}{\sigma - 1} \left[1 - \left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^{-\sigma} \right] \underbrace{\frac{\left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^\sigma (F_{hij0})^{-\alpha + \sigma - 1}}{1 + \left[\left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^\sigma - 1 \right]} \psi \frac{\beta}{1 - \beta} (1 - \omega_{hij}^{PRF})}_{\text{Long run share of PRF trade}} = \\ & -\frac{\alpha - \sigma + 1}{\sigma - 1} \left[1 - \left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^{-\sigma} \right] s_{hij}^{PRF} \psi \frac{\beta}{1 - \beta} (1 - \omega_{hij}^{PRF}) \end{aligned}$$

Substituting $1 - \omega_{hij}^{PRF}$ with (??) yields:

$$\frac{\partial \ln X_{hijt}}{\partial \gamma_t} = -\frac{\alpha - \sigma + 1}{\sigma - 1} \psi \frac{\beta}{1 - \beta} (1 - H(\tau_{hijt}^{PRF})) s_{hij}^{PRF} \left[1 - \left(\frac{\tau_{hij0}^{PRF}}{\tau_{hj0}^{MFN}} \right)^\sigma \right]$$

Note moreover that the derivation with respect to lagged uncertainty γ_{t-l} yields the same partial elasticity of current uncertainty γ_t .

Combining all the derivatives evaluated at the long run equilibrium with $\gamma = 0$ and rearranging we obtain:

$$\begin{aligned} \ln X_{hijt|t=0, \gamma=0} = & \ln C_{hij} + \frac{\alpha - \sigma + 1}{\sigma - 1} \psi \sum_{l=0}^{\infty} (1 - \psi)^l \ln \left(\frac{B_{jt-l}}{B_{j0}} \right) + \psi \sum_{l=0}^{\infty} (1 - \psi)^l \ln \left(\frac{m_{it-l}}{m_{i0}} \right) + \\ & \left[-\frac{\alpha \sigma}{\sigma - 1} \psi (1 - s_{hij}^{PRF}) - \sigma (1 - \psi) (1 - s_{hij}^{PRF}) \right] \ln \left(\frac{\tau_{hjt}^{MFN}}{\tau_{hj0}^{MFN}} \right) + \\ & \left[-\frac{\alpha \sigma}{\sigma - 1} \psi s_{hij}^{PRF} - \sigma (1 - \psi) s_{hij}^{PRF} \right] \ln \left(\frac{\tau_{hijt}^{PRF}}{\tau_{hij0}^{PRF}} \right) + \\ & -\sigma \frac{\alpha - \sigma + 1}{\sigma - 1} \psi \sum_{l=1}^{\infty} (1 - \psi)^l \left[s_{hij0}^{MFN} \ln \left(\frac{\tau_{hjt-l}^{MFN}}{\tau_{hj0}^{MFN}} \right) + s_{hij0}^{PRF} \ln \left(\frac{\tau_{hijt-l}^{PRF}}{\tau_{hij0}^{PRF}} \right) \right] + \\ & -\frac{\alpha - \sigma + 1}{\sigma - 1} \psi \frac{\beta}{1 - \beta} (1 - H(\tau_{hijt}^{PRF})) s_{hij0}^{PRF} \left[1 - \left(\frac{\tau_{hij0}^{PRF}}{\tau_{hj0}^{MFN}} \right)^\sigma \right] \sum_{l=0}^{\infty} (1 - \psi)^l \gamma_{t-l} \end{aligned}$$

Next

$$\begin{aligned}
\ln X_{hijt|t=0,\gamma=0} = & \ln C_{hij} + \frac{\alpha - \sigma + 1}{\sigma - 1} \psi \sum_{l=0}^{\infty} (1 - \psi)^l \ln \left(\frac{B_{jt-l}}{B_{j0}} \right) + \psi \sum_{l=0}^{\infty} (1 - \psi)^l \ln \left(\frac{m_{it-l}}{m_{i0}} \right) + \\
& - \sigma \left[1 + \frac{\alpha - \sigma + 1}{\sigma - 1} \psi \right] \left[s_{hij0}^{MFN} \ln \left(\frac{\tau_{hjt-l}^{MFN}}{\tau_{hj0}^{MFN}} \right) + s_{hij0}^{PRF} \ln \left(\frac{\tau_{hijt-l}^{PRF}}{\tau_{hij0}^{PRF}} \right) \right] \\
& - \sigma \frac{\alpha - \sigma + 1}{\sigma - 1} \psi \sum_{l=1}^{\infty} (1 - \psi)^l \left[s_{hij0}^{MFN} \ln \left(\frac{\tau_{hjt-l}^{MFN}}{\tau_{hj0}^{MFN}} \right) + s_{hij0}^{PRF} \ln \left(\frac{\tau_{hijt-l}^{PRF}}{\tau_{hij0}^{PRF}} \right) \right] + \\
& - \frac{\alpha - \sigma + 1}{\sigma - 1} \psi \frac{\beta}{1 - \beta} \left(1 - H(\tau_{hijt}^{PRF}) \right) s_{hij0}^{PRF} \left[1 - \left(\frac{\tau_{hij0}^{PRF}}{\tau_{hj0}^{MFN}} \right)^{\sigma} \right] \sum_{l=0}^{\infty} (1 - \psi)^l \gamma_{t-l}
\end{aligned}$$

Equilibrium share of preferential trade: The share of PRF trade in any period is computed as the ratio of PRF exports over total exports:

$$s_{hijt}^{PRF} = \left(\frac{\tau_{hjt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^{\sigma} \frac{\sum_{l=0}^{\infty} (1 - \psi)^l m_{it-l} \left(B_{jt-l} (\tau_{hjt-l}^{MFN})^{-\sigma} \right)^{\frac{\alpha - \sigma + 1}{\sigma - 1}} (F_{hijt-l} \times U_{t-l})^{-\alpha + \sigma - 1}}{\sum_{l=0}^{\infty} (1 - \psi)^l m_{it-l} \left(B_{jt-l} (\tau_{hjt-l}^{MFN})^{-\sigma} \right)^{\frac{\alpha - \sigma + 1}{\sigma - 1}} \left[1 + \left(\left(\frac{\tau_{hjt}^{MFN}}{\tau_{hijt}^{PRF}} \right)^{\sigma} - 1 \right) (F_{hijt-l} \times U_{t-l})^{-\alpha + \sigma - 1} \right]}$$

At any point in time the PRF share today also accounts for the history of shocks. At the steady state, the PRF share does not depends on its history. If tariff did not change for a period long enough – let's call this value 0 – we can take some terms out of the lag sums:

$$s_{hij0}^{PRF} = \left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^{\sigma} \frac{\left(\tau_{hj0}^{MFN} \right)^{-\sigma \frac{\alpha - \sigma + 1}{\sigma - 1}} (F_{hij0})^{-\alpha + \sigma - 1} \sum_{l=0}^{\infty} (1 - \psi)^l m_{it-l} (B_{jt-l})^{\frac{\alpha - \sigma + 1}{\sigma - 1}}}{\left(\tau_{hj0}^{MFN} \right)^{-\sigma \frac{\alpha - \sigma + 1}{\sigma - 1}} \left[1 + \left(\left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^{\sigma} - 1 \right) (F_{hij0})^{-\alpha + \sigma - 1} \right] \sum_{l=0}^{\infty} (1 - \psi)^l m_{it-l} (B_{jt-l})^{\frac{\alpha - \sigma + 1}{\sigma - 1}}}$$

Then the two lag sums cancel out and we are left with the long run share of PRF exports:

$$s_{hij0}^{PRF} = \frac{\left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^{\sigma} (F_{hij0})^{-\alpha + \sigma - 1}}{1 + \left(\left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^{\sigma} - 1 \right) (F_{hij0})^{-\alpha + \sigma - 1}} \quad (27)$$

Note also that we can invert (27) to have measure of $F_{hij0}^{-\alpha + \sigma - 1}$:

$$(F_{hij0})^{-\alpha + \sigma - 1} = \frac{s_{hij0}^{PRF}}{s_{hij0}^{PRF} + \left(1 - s_{hij0}^{PRF} \right) \left(\frac{\tau_{hj0}^{MFN}}{\tau_{hij0}^{PRF}} \right)^{\sigma}}$$

B.1 Tracking the evolution of uncertainty over time

In Table 4, I follow Graziano et al. (2021) and measure the evolution of uncertainty over time using the Brexit Uncertainty Index (BUI) – see Bloom et al. (2017). The index is computed from the Decision Maker Panel, a large survey of British firms, as the proportion of firms citing Brexit as a top-3 source of uncertainty. The regression model that uses the log of the BUI to track uncertainty over time is:

$$\begin{aligned} \ln X_{hit} = & a_{hi,month} + a_{it} + a_{ht} + \\ & \sum_{l=0}^3 b_l^\tau \overline{\ln \tau_{hit-l}} + \sum_{l=0}^L b_l^\gamma (\ln BUI_{t-l} \times s_{hi}^{PRF} \times \omega_{hi}) + \\ & \sum_{l=0}^L b_l^{CA} (\ln BUI_{t-l} \times s_{hi}^{PRF} \times \omega_{hi} \times CA_{it}) + e_{hit} \quad (28) \end{aligned}$$

I consider the current and up to two lags of the BUI, and compute the long-run elasticity to uncertainty as the sum of the coefficients b_l^γ . To account for zeros in the BUI, I take the hyperbolic sine transformation $\ln(x + (1 + x^2)^{1/2})$. For the Continuity Agreements, I interact the uncertainty measure with the dummy CA_{it} that takes a value of one if country i signed the continuity agreement in period t , for period t onwards. For the HL model, equation (28) is estimated setting $s_{hi}^{PRF} = 1$.

B.2 Comparison with the single tariff regime model

The HL model with a single trade regime between countries with a PTA gives the following equation for the long-run value of exports:

$$X^{HL} = A (\tau^{PRF})^{-\frac{\sigma\alpha}{\sigma-1}} \times \left[\frac{1 - \beta + \beta\gamma}{1 - \beta + \beta\gamma\omega^{HL}} \right]^{-\frac{\alpha-\sigma+1}{\sigma-1}}$$

where

$$\omega^{HL} = \frac{(1 - H(\tau^{PRF}))E(\tau^{-\sigma} | \tau' > \tau^{PRF}) + H(\tau^{PRF}) (\tau^{PRF})^{-\sigma}}{(\tau^{PRF})^{-\sigma}}$$

Omitting subscript for simplifying. Hence, comparing the uncertainty to the no-uncertainty scenarios while holding other variables constant gives:

$$\frac{X^{HL}(\gamma > 0)}{X^{HL}(\gamma = 0)} = \left[\frac{1 - \beta + \beta\gamma}{1 - \beta + \beta\gamma\omega^{HL}} \right]^{-\frac{\alpha-\sigma+1}{\sigma-1}} \quad (29)$$

The calculations of Section 6.2 are based on (29) setting $E(\tau^{-\sigma}|\tau' > \tau^{PRF}) = (\tau^{MFN})^{-\sigma}$.

C Data appendix

PTAs and Continuity Agreements information: Information on the PTA partners of the EU comes from the European Union commission website.¹⁶ Information on the UK Continuity Agreements come from the UK government website.¹⁷ For Continuity Agreements I identify two dates as the ones after which uncertainty could start to resolve to construct the CA_{ct} dummy variable in equation (12). First, I review the official CA documents as published by the UK government and press releases to find the dates in which the CAs were signed by each country. For instance, for the CARIFORUM countries the CA, which was presented to Parliament on 22/05/2019, nine out of fourteen countries signed on 22/03/2019 while the remaining five countries signed later on. The signature date is used to construct the CA_{ct} dummy, and results are checked against using the date of presentation to the UK Parliament. Using the month of presentation to Parliament of the CA could introduce some measurement error in the CA_{ct} variable. When a CA covers more than one country, as for the CARIFORUM trade bloc, it is not always the case that all countries covered by the EU PTA signed the CA with the UK in the same period. Moreover, in some instances the CA was presented to Parliament a few months after it had been signed. If the main signal for the reduction of uncertainty was the signature of the deal, then the presentation to Parliament date might consider actually treated periods as non-treated by the CAs, lowering the chances of finding any significant CA effect in the regressions.

For both the signature and presentation to Parliament dates, if the CA date lies in the first half of month t then I set $CA_{ct} = 1$ from month t onwards, while if the CA date lies in the second half of month t I set $CA_{ct} = 1$ from month $t + 1$ onwards.

In total, out of the 70 PTA partners considered, 48 of them signed a Continuity Agreement with the UK in 2019 and to date, only 4 countries did not sign a CA at all.¹⁸ By the end of 2019 the UK signed

¹⁶See <https://ec.europa.eu/trade/policy/countries-and-regions/negotiations-and-agreements/>. We consider the agreements in place as per December 2021.

¹⁷See <https://www.gov.uk/guidance/uk-trade-agreements-with-non-eu-countries>

¹⁸Of of the 70 considered, the actual number of countries without a CA with the UK is 6. However, this includes Andorra and San Marino which are covered by the Trade and Cooperation Agreement (TCA) between the UK and the EU. As reported in the UK government website, under the TCA products originating in Andorra and San Marino ‘are to be treated as originating in the EU under the UK-EU Trade and Cooperation Agreement as long as Andorra and San Marino apply to UK products the same preferences that the EU applies to them. These products benefit from zero tariffs, where they meet the relevant rules of origin. Trade with Andorra of agricultural products (chapters 01-24) is not in scope of the UK-EU Trade and Cooperation Agreement and takes place under WTO terms.’ See: <https://www.gov.uk/guidance/uk-trade-agreements-with-non-eu-countries>. The four countries that did not sign a continuity agreement are Armenia, Bosnia and Herzegovina, Comoros and Montenegro.

CAs for a substantial portion of its imports from PTA partners. The two largest ‘signature months’ by imports volume are February 2019, when Switzerland signed, and April 2019 when Iceland and Norway signed.

UK imports data: The UK imports dataset includes all EU PTA partners which enjoyed preferential access in 2015 and that now have a trade agreement in place with the EU.¹⁹ Some of these countries had preferential access to the EU market before 2016 under the GSP or EBA schemes, and signed an trade agreement only after 2015, while other countries such as South Africa changed the type of agreement in the period 2015-19. I therefore consider both the full sample, which counts 70 countries, and the restricted sample of countries with PTAs in place before 2013 and that did not change agreement in the period 2013-19. This restricted sample counts 41 countries.

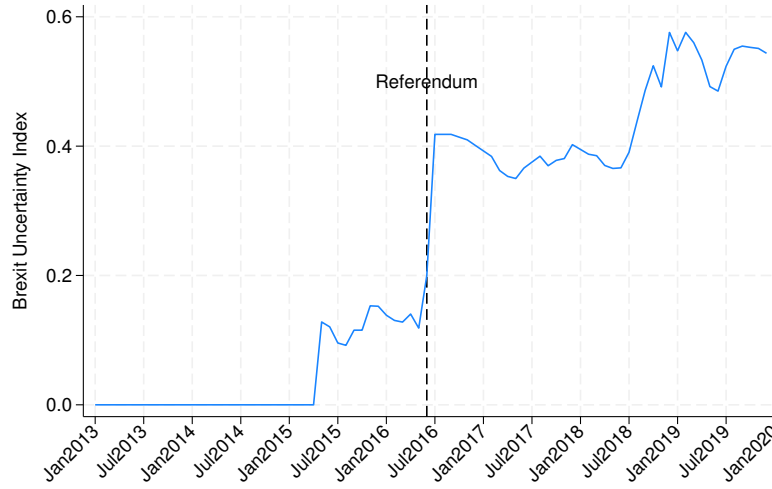
The analysis is carried out at the CN 8-digit product level with monthly frequency using data for 2013-19. Given changes in the CN classification over time, I use the concordance procedure of Van Beveren et al. (2012).

Data for UK imports from the PTA partners and preference utilisation rates come from the Eurostat Comext database. I use data at the CN 8-digit level with monthly frequency for 2013-19. I exclude imports for inward and outward processing (codes 2 and 3 of the Eurostat Comext statistical regime). These flows concern processing trade and products imported under these regimes are fully or partially exempt from customs duties. Since no duty is levied on these flows, they are not subject from tariff uncertainty.

Data on preference utilisation are available for all EU PTA partner countries at monthly frequency and they cover the totality of EU imports value at the CN 8-digit product level. The PREF dataset gives information on the imports by country, CN 8-digit product, period, statistical regime (Normal trade, processing, not for customs declarations) and import regime. The information required for the calculation of the PRF import share is the import regime. The Eurostat dataset consider the MFN regime, preferential and GSP regimes, and a residual ‘unknown’ category. The ‘unknown’ regime accounts for a small fraction of imports: on average over 2013-15, only 3% of total UK imports are classed under the ‘unknown’ import regime. In the calculation of the PRF share, I drop imports under ‘unknown’ regime and aggregate the GSP and preferential regimes into a single ‘non-MFN’ regime for the calculation of the PRF import shares. The baseline PRF shares are computed with data for 2013-15. This is done adding imports over time at the partner-product-regime level (=MFN, PRF), and then compute the PRF share on total imports by partner-product over 2013-15.

¹⁹Of the current EU PTA partners, I exclude Canada, Japan, Singapore and Vietnam as they signed trade agreements with the EU after 2015.

Figure 6: Brexit Uncertainty Index



Source: author's elaboration of the Brexit Uncertainty Index.

Tariff data: MFN and preferential tariff data for the years 2010-19 come from the ITC MacMap database for all countries at the tariff-line level and annual frequency. They are then merged with imports data at the partner-CN8-year level, and they include current and lagged (up to 3 years) tariffs.

Brexit Uncertainty Index: The Brexit Uncertainty Index (BUI) is taken from the Bank of England Decision Maker Panel. This is a large survey of British firms asking what are the major sources of uncertainty, and contains Brexit-specific questions. The index is computed as the proportion of firms citing Brexit as a top-3 source of uncertainty. As the survey started after the referendum in September 2016, the authors of the BUI extrapolated values for the index from August 2016 to May 2015 using betting odds data, and for periods before May 2015 the index takes value of zero. Figure 6 plots the index over the period Jan2013-Dec2019.

Table 5: PTA and Continuity Agreements

Country	PTA year	CA Parliament	CA signature	Country	PTA year	CA Parliament	CA signature
Albania	2009	19/02/2021	05/02/2021	El Salvador	2013	06/08/2019	18/07/2019
Algeria	2005			Eswatini	2016	05/11/2019	09/10/2019
Andorra	1991			Faroe Islands	1997	06/02/2019	31/01/2019
Antigua and Barbuda	2008	22/05/2019	06/06/2019	Fiji	2014	20/03/2019	14/03/2019
Armenia	2018			Georgia	2016	04/11/2019	21/10/2019
Bahamas	2008	22/05/2019	27/11/2019	Ghana	2016	20/04/2021	02/03/2021
Barbados	2008	22/05/2019	22/03/2019	Grenada	2008	22/05/2019	22/03/2019
Belize	2008	22/05/2019	22/03/2019	Guatemala	2013	06/08/2019	18/07/2019
Bosnia and Herzegovina	2015			Guyana	2008	22/05/2019	22/03/2019
Botswana	2016	05/11/2019	09/10/2019	Honduras	2013	06/08/2019	18/07/2019
Cameroon	2013	20/04/2021	09/03/2021	Iceland	1994	12/04/2019	02/04/2019
Chile	2003	06/02/2019	30/01/2019	Israel	2000	26/02/2019	18/02/2019
Colombia	2013	14/06/2019	15/05/2019	Jamaica	2008	22/05/2019	22/03/2019
Comoros	2019			Jordan	2002	20/12/2019	05/11/2019
Costa Rica	2013	06/08/2019	18/07/2019	Kosovo	2016	20/12/2019	03/12/2019
Cote d'Ivoire	2016	11/11/2020	15/10/2020	Lebanon	2006	22/10/2019	19/09/2019
Dominica	2008	22/05/2019	22/03/2019	Lesotho	2016	05/11/2019	09/10/2019
Dominican Republic	2008	22/05/2019	04/04/2019	Liechtenstein	1995	28/02/2019	11/02/2019
Ecuador	2017	14/06/2019	15/05/2019	Madagascar	2012		
Egypt	2004	14/12/2020	05/12/2020	Mauritius	2012	06/02/2019	31/01/2019

Table 6: PTA and Continuity Agreements continued

Country	PTA year	CA Parliament	CA signature	Country	PTA year	CA Parliament	CA signature
Mexico	2000	26/02/2021	15/12/2020	Serbia	2013	11/05/2021	16/04/2021
Moldova	2016	18/01/2021	24/12/2020	Seychelles	2012	06/02/2019	31/01/2019
Montenegro	2010			Solomon Islands	2020	20/03/2019	22/12/2020
Morocco	2000	20/12/2019	26/10/2019	South Africa	1999	05/11/2019	09/10/2019
Mozambique	2016	05/11/2019	09/10/2019	South Korea	2011	10/09/2019	22/08/2019
Namibia	2016	05/11/2019	09/10/2019	St Kitts and Nevis	2008	22/05/2019	22/03/2019
Nicaragua	2013	06/08/2019	18/07/2019	St Lucia	2008	22/05/2019	22/03/2019
North Macedonia	2004	10/12/2020	03/12/2020	St Vincent and the Grenadines	2008	22/05/2019	22/03/2019
Norway	1994	12/04/2019	02/04/2019	Suriname	2008	22/05/2019	04/03/2021
Palestinian Authority	1997	26/02/2019	18/02/2019	Switzerland	1973	20/02/2019	11/02/2019
Panama	2013	06/08/2019	18/07/2019	Trinidad and Tobago	2008	22/05/2019	01/04/2019
Papua New Guinea	2011	20/03/2019	14/03/2019	Tunisia	1998	25/10/2019	04/10/2019
Peru	2013	14/06/2019	15/05/2019	Turkey	1995	24/02/2021	29/12/2020
Samoa	2018	20/03/2019	22/12/2020	Ukraine	2016	09/11/2020	08/10/2020
San Marino	1991			Zimbabwe	2012	06/02/2019	31/01/2019

D Additional results

D.1 IV regressions

This section reports the IV regressions without dealing with staggered adoption and estimating one uncertainty coefficient for each year rather than for each year-month. For Continuity Agreements, I interact the MFN risk with a dummy that equals one if the country signed a CA in period t and zero otherwise.

Table 7 reports the second-stage regressions. In all cases we can see that F-stats are very large. In column 1 I instrument the 2013-15 UK PRF share s_{hi}^{PRF} with the median PRF share of the EU27 computed

Table 7: IV regression, second stage

	Instrumented variable			
	(1) PRF share 13-15	(2) Tariff margin	(3) PRF share 13-19	(4) shr 13-19; tariff mrg
Current tariff	-2.263*** (0.757)	-1.532** (0.682)	-2.448*** (0.742)	-1.929** (0.750)
Tariff, 1-year lag	-1.539** (0.688)	-0.813 (0.623)	-1.645** (0.662)	-1.034 (0.669)
Tariff, 2-year lag	0.430 (0.620)	0.236 (0.543)	0.331 (0.594)	0.497 (0.598)
Tariff, 3-year lag	0.012 (0.486)	-0.056 (0.408)	0.013 (0.457)	0.074 (0.459)
MFN risk 2016	-0.405** (0.160)	-0.447*** (0.121)	-0.431*** (0.159)	-0.969*** (0.213)
MFN risk 2017	-0.422*** (0.158)	-0.877*** (0.120)	-0.425*** (0.155)	-1.033*** (0.206)
MFN risk 2018	-0.926*** (0.159)	-1.486*** (0.122)	-0.852*** (0.155)	-1.700*** (0.208)
MFN risk 2019	-1.182*** (0.164)	-1.598*** (0.129)	-1.223*** (0.161)	-1.910*** (0.211)
Continuity Agreement	0.489*** (0.109)	0.751*** (0.128)	0.472*** (0.107)	0.952*** (0.145)
Partner-product-month FE	Yes	Yes	Yes	Yes
Partner-time FE	Yes	Yes	Yes	Yes
Product-time FE	Yes	Yes	Yes	Yes
Observations	545,938	549,992	554,734	554,734
F first-stage	21,095.9	80,648.7	26,701.5	10,710.8

Standard errors clustered at partner-CN8 level in parenthesis. in parenthesis. * p<0.1, ** p<0.05, *** p<0.01. Column 1 instruments the UK PRF share 2013-15 with the EU PRF share 2013-19. Column 2 instruments the tariff margin with the median margin computed with MFN tariffs of Australia, Canada, Japan and the US. Column 3 instruments the UK PRF share 2013-19 with the EU share 2013-19. Column 4 instruments both the UK share 2013-19 and the tariff margin. The dependent variable is the log of UK imports from PTA partners excluding processing trade.

over 2013-19. In column 2 I instrument the tail risk measure $\omega_{hi} = 1 - (\tau_{hi}^{PRF} / \tau_h^{MFN})^\sigma$ using the median tail risk computed with the MFN tariffs of Australia, Canada, Japan and the US. In column 3 I instrument the PRF import share of the UK over 2013-19 with the EU PRF share for the same period to maximise sample coverage. Finally, in column 4 I instrument both the 2013-19 UK PRF import share and the tail risk ω_{hi} .

The first-stage regressions are reported in Tables 8-11. Each columns represents a first-stage for a given endogenous variable and the relevant instrument can be found on the diagonal. We can see that the relevant instruments are always strongly significant and have coefficients close to unity.

Table 8: IV PRF share 2013-15, first stage

	Tariff				MFN risk				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	lag=0	lag=1	lag=2	lag=3	2016	2017	2018	2019	Cont. Ag.
Tariff IV, lag=0	0.908*** (0.001)	-0.038*** (0.001)	-0.015*** (0.001)	-0.016*** (0.002)	0.045*** (0.011)	0.097*** (0.011)	-0.003 (0.011)	0.050*** (0.011)	0.036*** (0.007)
Tariff IV, lag=1	-0.022*** (0.001)	0.904*** (0.001)	-0.046*** (0.001)	-0.027*** (0.002)	0.100*** (0.010)	0.078*** (0.010)	0.084*** (0.010)	0.006 (0.010)	0.002 (0.006)
Tariff IV, lag=2	-0.005*** (0.001)	-0.032*** (0.001)	0.895*** (0.001)	-0.015*** (0.001)	0.041*** (0.009)	0.051*** (0.009)	0.040*** (0.009)	0.081*** (0.009)	0.041*** (0.006)
Tariff IV, lag=3	-0.000 (0.000)	-0.008*** (0.001)	-0.016*** (0.001)	0.861*** (0.001)	-0.077*** (0.007)	0.033*** (0.007)	0.097*** (0.007)	0.117*** (0.007)	0.042*** (0.004)
MFN risk IV, 2016	-0.002*** (0.000)	0.000*** (0.000)	0.002*** (0.000)	0.003*** (0.000)	0.717*** (0.002)	-0.007*** (0.002)	-0.003* (0.002)	-0.004** (0.002)	-0.002* (0.001)
MFN risk IV, 2017	-0.002*** (0.000)	-0.000* (0.000)	0.001*** (0.000)	0.002*** (0.000)	-0.012*** (0.002)	0.726*** (0.002)	-0.003 (0.002)	-0.005*** (0.002)	-0.003** (0.001)
MFN risk IV, 2018	-0.003*** (0.000)	-0.001*** (0.000)	0.000** (0.000)	0.004*** (0.000)	-0.012*** (0.002)	-0.009*** (0.002)	0.731*** (0.002)	-0.006*** (0.002)	-0.002** (0.001)
MFN risk IV, 2019	-0.003*** (0.000)	-0.002*** (0.000)	-0.000 (0.000)	0.004*** (0.000)	-0.013*** (0.002)	-0.012*** (0.002)	-0.008*** (0.002)	0.740*** (0.002)	-0.086*** (0.001)
Cont. Ag. IV	0.000*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.001*** (0.000)	0.001 (0.002)	0.002 (0.002)	0.002 (0.002)	-0.008*** (0.002)	0.939*** (0.001)
Observations	545,938	545,938	545,938	545,938	545,938	545,938	545,938	545,938	545,938
Partner-CN8-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Partner-time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
CN8-time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors clustered at partner-CN8 level in parenthesis. * p<0.1, ** p<0.05, *** p<0.01. The table reports the first stages using the EU PRF share 2013-15 to instrument the UK PRF share 2013-15. In columns 1-4 the dependent variable is the average applied tariff computed as $\ln \tau_{hit} = s_{hi}^{PRF} \ln \tau_{hit}^{PRF} + s_{hi}^{MFN} \ln \tau_{hit}^{MFN}$. In columns 5-8 the dependent variable is the MFN risk computed as $s_{hi}^{PRF} \times \omega_{hi}$, where $\omega = 1 - (\tau^{PRF}/\tau^{MFN})^4$. In column 9 the dependent variable is the interaction of the Continuity Agreement dummy with the MFN risk.

Table 9: IV tariff margin, first stage

	(1)	(2)	(3)	(4)	(5)
	MFN risk, 2016	MFN risk, 2017	MFN risk, 2018	MFN risk, 2019	Cont. Ag.
MFN risk IV, 2016	1.102*** (0.002)	-0.000 (0.002)	0.001 (0.002)	-0.002 (0.002)	-0.002 (0.002)
MFN risk IV, 2017	-0.001 (0.002)	1.113*** (0.002)	-0.003* (0.002)	-0.004*** (0.002)	-0.001 (0.002)
MFN risk IV, 2018	0.000 (0.002)	-0.004** (0.002)	1.117*** (0.002)	-0.006*** (0.002)	-0.003* (0.002)
MFN risk IV, 2019	0.000 (0.002)	-0.001 (0.002)	-0.003* (0.002)	1.110*** (0.002)	0.060*** (0.002)
Continuity Agreement IV	0.001 (0.001)	0.004*** (0.001)	0.003* (0.001)	-0.012*** (0.001)	0.955*** (0.002)
Tariff, lag=0	0.294*** (0.008)	0.229*** (0.008)	-0.025*** (0.008)	0.245*** (0.008)	0.170*** (0.009)
Tariff, lag=2	0.121*** (0.007)	0.273*** (0.007)	0.261*** (0.007)	-0.001 (0.007)	0.014* (0.008)
Tariff, lag=2	-0.345*** (0.006)	0.034*** (0.006)	0.166*** (0.006)	0.212*** (0.006)	0.155*** (0.007)
Tariff, lag=3	0.017*** (0.005)	-0.219*** (0.005)	-0.008* (0.005)	0.103*** (0.005)	0.029*** (0.005)
Observations	549,992	549,992	549,992	549,992	549,992
Partner-CN8-month FE	Yes	Yes	Yes	Yes	Yes
Partner-time FE	Yes	Yes	Yes	Yes	Yes
CN8-time FE	Yes	Yes	Yes	Yes	Yes

Standard errors clustered at partner-CN8 level in parenthesis.in parenthesis. * p<0.1, ** p<0.05, *** p<0.01. The table reports the first stages using median of the tail risk computed with the MFN tariffs of the Australia, Canada, Japan and the US to instrument the UK tail risk. In columns 1-4 the dependent variable is the MFN risk computed as $s_{hi}^{PRF} \times \omega_{hi}$, where $\omega = 1 - (\tau^{PRF} / \tau^{MFN})^4$. In column 5 the dependent variable is the interaction of the Continuity Agreement dummy with the MFN risk.

Table 10: IV PRF share 2013-19, first stage

	Tariff				MFN risk				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	lag=0	lag=1	lag=2	lag=3	2016	2017	2018	2019	Cont. Ag.
Tariff IV, lag=0	0.922*** (0.001)	-0.029*** (0.001)	-0.010*** (0.001)	-0.010*** (0.002)	0.042*** (0.010)	0.093*** (0.010)	-0.016 (0.010)	0.040*** (0.009)	0.022*** (0.006)
Tariff IV, lag=1	-0.020*** (0.001)	0.931*** (0.001)	-0.032*** (0.001)	-0.014*** (0.001)	0.083*** (0.009)	0.061*** (0.009)	0.080*** (0.009)	0.001 (0.009)	0.002 (0.005)
Tariff IV, lag=2	-0.002*** (0.001)	-0.021*** (0.001)	0.932*** (0.001)	-0.006*** (0.001)	-0.011 (0.008)	0.043*** (0.008)	0.028*** (0.008)	0.080*** (0.008)	0.035*** (0.005)
Tariff IV, lag=3	0.002*** (0.000)	-0.001* (0.000)	-0.006*** (0.001)	0.909*** (0.001)	-0.034*** (0.006)	-0.010 (0.006)	0.073*** (0.006)	0.077*** (0.006)	0.026*** (0.004)
MFN risk IV, 2016	-0.002*** (0.000)	0.000*** (0.000)	0.002*** (0.000)	0.003*** (0.000)	0.712*** (0.002)	-0.004*** (0.002)	-0.001 (0.002)	-0.003* (0.002)	-0.002** (0.001)
MFN risk IV, 2017	-0.001*** (0.000)	-0.001*** (0.000)	0.001*** (0.000)	0.003*** (0.000)	-0.010*** (0.002)	0.730*** (0.002)	-0.003* (0.002)	-0.005*** (0.002)	-0.002** (0.001)
MFN risk IV, 2018	-0.002*** (0.000)	-0.001*** (0.000)	0.000 (0.000)	0.004*** (0.000)	-0.009*** (0.002)	-0.006*** (0.002)	0.731*** (0.002)	-0.006*** (0.002)	-0.004*** (0.001)
MFN risk IV, 2019	-0.002*** (0.000)	-0.001*** (0.000)	0.000 (0.000)	0.003*** (0.000)	-0.012*** (0.002)	-0.009*** (0.002)	-0.008*** (0.002)	0.744*** (0.002)	-0.078*** (0.001)
Cont. Ag. IV	0.001*** (0.000)	0.001*** (0.000)	0.000* (0.000)	-0.000* (0.000)	0.003** (0.001)	0.005*** (0.001)	0.003* (0.001)	-0.021*** (0.001)	0.939*** (0.001)
Observations	554,734	554,734	554,734	554,734	554,734	554,734	554,734	554,734	554,734
Partner-CN8-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Partner-time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
CN8-time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors clustered at partner-CN8 level in parenthesis. * p<0.1, ** p<0.05, *** p<0.01. The table reports the first stages using the EU PRF share 2013-19 to instrument the UK PRF share 2013-19. In columns 1-4 the dependent variable is the average applied tariff computed as $\ln \tau_{hit} = s_{hi}^{PRF} \ln \tau_{hit}^{PRF} + s_{hi}^{MFN} \ln \tau_{ht}^{MFN}$. In columns 5-8 the dependent variable is the MFN risk computed as $s_{hi}^{PRF} \times \omega_{hi}$, where $\omega = 1 - (\tau^{PRF}/\tau^{MFN})^4$. In column 9 the dependent variable is the interaction of the Continuity Agreement dummy with the MFN risk.

Table 11: IV PRF share 2013-19 and tariff margin, first stage

	Tariff				MFN risk				
	(1) lag=0	(2) lag=1	(3) lag=2	(4) lag=3	(5) 2016	(6) 2017	(7) 2018	(8) 2019	(9) Cont. Ag.
Tariff IV, lag=0	0.920*** (0.001)	-0.030*** (0.001)	-0.009*** (0.001)	-0.009*** (0.002)	0.234*** (0.011)	0.271*** (0.011)	-0.034*** (0.011)	0.187*** (0.011)	0.123*** (0.010)
Tariff IV, lag=1	-0.021*** (0.001)	0.931*** (0.001)	-0.031*** (0.001)	-0.012*** (0.001)	0.188*** (0.010)	0.221*** (0.010)	0.269*** (0.010)	-0.039*** (0.010)	-0.021** (0.009)
Tariff IV, lag=2	-0.002*** (0.001)	-0.021*** (0.001)	0.931*** (0.001)	-0.007*** (0.001)	-0.299*** (0.009)	0.067*** (0.009)	0.109*** (0.009)	0.218*** (0.009)	0.156*** (0.008)
Tariff IV, lag=3	0.002*** (0.000)	-0.001* (0.000)	-0.007*** (0.001)	0.909*** (0.001)	0.001 (0.007)	-0.181*** (0.007)	0.064*** (0.007)	0.152*** (0.007)	0.044*** (0.006)
MFN risk IV, 2016	-0.001*** (0.000)	0.000 (0.000)	0.002*** (0.000)	0.006*** (0.000)	0.715*** (0.002)	-0.010*** (0.003)	-0.006** (0.002)	-0.010*** (0.002)	-0.007*** (0.002)
MFN risk IV, 2017	-0.001*** (0.000)	-0.000 (0.000)	0.001*** (0.000)	0.006*** (0.000)	-0.012*** (0.002)	0.736*** (0.002)	-0.014*** (0.002)	-0.015*** (0.002)	-0.007*** (0.002)
MFN risk IV, 2018	-0.001*** (0.000)	-0.000** (0.000)	0.001*** (0.000)	0.007*** (0.000)	-0.008*** (0.002)	-0.015*** (0.002)	0.738*** (0.002)	-0.019*** (0.002)	-0.012*** (0.002)
MFN risk IV, 2019	-0.001*** (0.000)	-0.000 (0.000)	0.000** (0.000)	0.006*** (0.000)	-0.014*** (0.003)	-0.016*** (0.003)	-0.020*** (0.003)	0.755*** (0.003)	-0.062*** (0.002)
Cont. Ag. IV	0.000** (0.000)	0.000** (0.000)	0.000 (0.000)	-0.000 (0.000)	0.009*** (0.002)	0.014*** (0.002)	0.013*** (0.002)	-0.063*** (0.002)	0.872*** (0.002)
Observations	554,734	554,734	554,734	554,734	554,734	554,734	554,734	554,734	554,734
Partner-CN8-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Partner-time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
CN8-time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors clustered at partner-CN8 level in parenthesis. * p<0.1, ** p<0.05, *** p<0.01. The table reports the first stages using the EU PRF share 2013-19 to instrument the UK PRF share 2013-19 and the median of the tail risk computed with the MFN tariffs of the Australia, Canada, Japan and the US to instrument the UK tail risk. In columns 1-4 the dependent variable is the average applied tariff computed as $\ln \tau_{hit} = s_{hi}^{PRF} \ln \tau_{hit}^{PRF} + s_{hi}^{MFN} \ln \tau_{hit}^{MFN}$. In columns 5-8 the dependent variable is the MFN risk computed as $s_{hi}^{PRF} \times \omega_{hi}$, where $\omega = 1 - (\tau^{PRF} / \tau^{MFN})^4$. In column 9 the dependent variable is the interaction of the Continuity Agreement dummy with the MFN risk.

Table 12: Variation in variables across CN8 products and countries

Variation (sd)	ω_{hi}	s_{hi}^{PRF}	$\omega_{hi} \times s_{hi}^{PRF}$
Overall	0.137	0.454	0.129
Between CN8	0.139	0.366	0.120
Within CN8	0.037	0.328	0.078

D.2 Other results

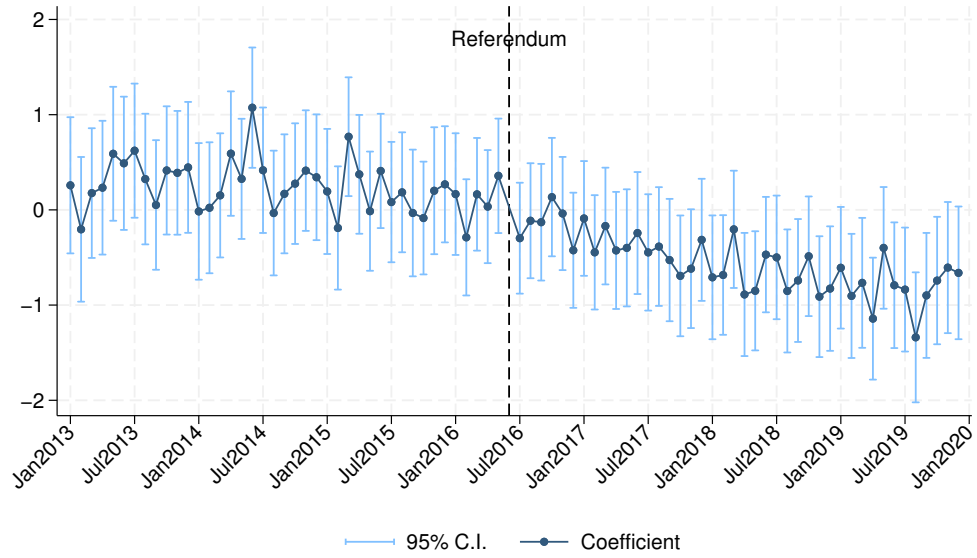
To visually inspect whether there are any pre-referendum trends that might bias the analysis I use an event-study regression:

$$\ln X_{hit} = a_{hi} + a_{ht} + a_{it} + \sum_{t \neq \text{Jun2016}} b_t (\omega_{hi} \times s_{hi}^{PRF} \times a_t) + e_{hit} \quad (30)$$

where a_{hi} , a_{ht} and a_{it} are CN8-partner, CN8-time and partner-time fixed effects, and the a_t s are time dummies. The coefficients on the uncertainty measure $\omega_{hi} \times s_{hi}^{PRF}$ are estimated relative to June 2016, the month in which the referendum was held. Compared to the baseline specification of equation (12), the regression model in (30) does not deal with seasonality, but it allows me to see whether we should worry about pre-treatment trends and it is useful to see how the uncertainty effect varied over time.

The coefficient estimates with the 95% confidence interval are plotted in Figure 7. Overall, there is no sign of pre-referendum trends. On the other hand, I find consistent negative coefficients for the post-referendum period spanning from July 2016 to December 2019. Moreover, the uncertainty effects have been growing over time in line with the presence of legacy firms and the idea that uncertainty effects build up over time.

Figure 7: Event study plot, UK imports from PTA partners



The figure plots the coefficients of the event-study regression with their 95% confidence interval. For the event-study we regress the log of UK imports from PTA partners at the CN 8-digit level over 2013-19 on the partner-product measure of uncertainty $\omega \times s^{PRF}$ interacted with time dummies. The reference period is June 2016. The regression includes partner-product, partner-time and product-time fixed effects. Standard errors are clustered at the partner-product level.

Table 13: Robustness tests

	(1) Pre-2013	(2) UK-EU	(3) Ex synth prod	(4) Processing
<i>Tariff</i>				
current	-3.197** (1.433)	-2.881* (1.510)	-1.854 (1.505)	-1.272 (0.980)
1-year lag	-1.450 (1.090)	-0.738 (1.489)	-0.347 (1.094)	-0.749 (0.742)
2-year lag	0.566 (0.976)	-0.565 (1.010)	-0.680 (0.979)	0.014 (0.713)
3-year lag	-0.549 (0.878)	-0.684 (0.895)	-0.174 (0.847)	-0.104 (0.644)
<i>Uncertainty effects, yearly average</i>				
2016	-0.365* (0.189)	-0.251 (0.223)	-0.326** (0.164)	-0.203 (0.140)
2017	-0.846*** (0.225)	-0.619** (0.247)	-0.733*** (0.195)	-0.473*** (0.171)
2018	-1.245*** (0.241)	-0.969*** (0.268)	-1.165*** (0.211)	-0.932*** (0.185)
2019	-1.350*** (0.270)	-1.041*** (0.290)	-1.503*** (0.238)	-1.233*** (0.206)
<i>Continuity Agreements, average Feb-Dec 2019:</i>				
Simple average	0.462** (0.195)	0.043 (0.242)	0.528*** (0.198)	0.355** (0.167)
Weighted average	0.544*** (0.192)	-0.102 (0.233)	0.566*** (0.196)	0.400** (0.162)
<i>Test if Continuity Agreement + Uncertainty effects = 0 over Feb-Dec 2019</i>				
Simple average	-0.944*** (0.287)	-1.010*** (0.326)	-1.035*** (0.266)	-0.928*** (0.223)
Weighted average	-0.862*** (0.286)	-1.155*** (0.325)	-0.998*** (0.270)	-0.883*** (0.224)
Observations	445,367	486,072	460,138	550,514

Table 14: HL model, different product-time fixed effects

	(1)	(2)	(3)	(4)	(5)
Current tariff	-0.987** (0.446)	-0.904* (0.476)	-1.118** (0.562)	-1.226* (0.689)	-1.764** (0.787)
Tariff, 1-year lag	-0.736** (0.359)	-0.671* (0.377)	-0.904** (0.394)	-1.125** (0.473)	-1.804*** (0.647)
Tariff, 2-year lag	-0.375 (0.330)	0.067 (0.338)	-0.010 (0.424)	-0.316 (0.516)	0.067 (0.649)
Tariff, 3-year lag	-0.180 (0.311)	-0.063 (0.314)	-0.264 (0.390)	-0.242 (0.490)	-0.285 (0.570)
Uncertainty 2016	-0.047 (0.053)	0.129 (0.083)	0.039 (0.113)	0.094 (0.154)	0.181 (0.232)
Uncertainty 2017	-0.182*** (0.063)	0.074 (0.096)	0.039 (0.133)	0.232 (0.177)	0.542* (0.291)
Uncertainty 2018	-0.345*** (0.070)	0.031 (0.108)	0.054 (0.150)	0.159 (0.206)	0.329 (0.357)
Uncertainty 2019	-0.478*** (0.086)	-0.254** (0.127)	-0.247 (0.175)	-0.161 (0.241)	0.072 (0.442)
Continuity Agreement	0.449*** (0.099)	0.395*** (0.105)	0.406*** (0.114)	0.377*** (0.127)	0.414*** (0.137)
Observations	705,284	705,180	680,610	609,345	549,537
Partner-CN8-month FE	Yes	Yes	Yes	Yes	Yes
Partner-time FE	Yes	Yes	Yes	Yes	Yes
Product-time FE		HS2	HS4	HS6	CN8

Standard errors clustered at partner-CN8 level in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table reports estimates for the Handley Limao model with fixed effects at different product-time level. Column 1 does not include product-time FE. For the other columns the levels are: column 2 at the HS 2-digit, column 3 at the HS 4-digit, column 4 at the HS 6-digit and column 5 at CN 8-digit level. The dependent variable is the log of UK imports from PTA partners excluding processing trade. The tariff is the preferential tariff. Uncertainty coefficients are estimated for each year between 2016-2019. The CA effect is computed interacting the uncertainty measure with a dummy for the Continuity Agreements. Uncertainty is measured as $\omega = 1 - (\tau^{PRF} / \tau^{MFN})^4$.

Table 15: PRF model, different product-time fixed effects

	(1)	(2)	(3)	(4)	(5)
Current tariff	-1.285*** (0.491)	-1.111** (0.517)	-1.692*** (0.652)	-1.740** (0.852)	-1.811* (0.997)
Tariff, 1-year lag	-0.678* (0.366)	-0.625* (0.377)	-0.852** (0.399)	-0.741 (0.487)	-1.260* (0.756)
Tariff, 2-year lag	-0.506* (0.303)	-0.126 (0.299)	-0.065 (0.362)	-0.246 (0.441)	0.112 (0.720)
Tariff, 3-year lag	-0.167 (0.320)	-0.026 (0.323)	-0.157 (0.411)	0.128 (0.500)	-0.127 (0.649)
Uncertainty 2016	-0.112** (0.055)	-0.068 (0.076)	-0.174* (0.095)	-0.136 (0.118)	-0.258* (0.147)
Uncertainty 2017	-0.315*** (0.065)	-0.275*** (0.089)	-0.402*** (0.111)	-0.371*** (0.136)	-0.657*** (0.177)
Uncertainty 2018	-0.494*** (0.073)	-0.382*** (0.100)	-0.538*** (0.121)	-0.615*** (0.150)	-1.039*** (0.190)
Uncertainty 2019	-0.669*** (0.091)	-0.715*** (0.120)	-0.862*** (0.144)	-0.906*** (0.177)	-1.229*** (0.211)
Continuity Agreement	0.413*** (0.110)	0.383*** (0.115)	0.429*** (0.125)	0.380*** (0.139)	0.386** (0.150)
Observations	693,985	693,881	669,588	599,264	540,302
Partner-CN8-month FE	Yes	Yes	Yes	Yes	Yes
Partner-time FE	Yes	Yes	Yes	Yes	Yes
Product-time FE		HS2	HS4	HS6	CN8

Standard errors clustered at partner-CN8 level in parenthesis.in parenthesis. * p<0.1, ** p<0.05, *** p<0.01. The table reports estimates of the PRF model with fixed effects at different product-time level. Column 1 does not include product-time FE. For the other columns the levels are: column 2 at the HS 2-digit, column 3 at the HS 4-digit, column 4 at the HS 6-digit and column 5 at CN 8-digit level. The dependent variable is the log of UK imports from PTA partners excluding processing trade (full sample). The tariff is the weighted-average applied tariff $\ln \tau_{hit} = s_{hi}^{PRF} \ln \tau_{hit}^{PRF} + s_{hi}^{MFN} \ln \tau_{hit}^{MFN}$. Uncertainty coefficients are estimated for each year between 2016-2019. The CA effect is computed interacting the uncertainty measure with a dummy for the Continuity Agreements. Uncertainty is measured as $s_{hi}^{PRF} \times \omega_{hi}$, where $\omega = 1 - (\tau^{PRF} / \tau^{MFN})^4$ and s^{PRF} is the share of PRF imports over 2013-15.

Table 16: GHL sample, HL model and Brexit Uncertainty Index

	(1)	(2)	(3)	(4)	(5)
Uncertainty $\times \ln BUI_t$	-1.059*** (0.196)	-0.384 (0.313)	-0.243 (0.492)	-0.328 (0.698)	1.957 (1.934)
CA $\times$ Uncertainty $\times \ln BUI_t$	0.930*** (0.200)	0.837*** (0.239)	1.094*** (0.281)	0.998*** (0.327)	1.268*** (0.371)
Tariffs, lags 0-3	Yes	Yes	Yes	Yes	Yes
Observations	519,591	519,146	492,469	418,990	360,092
Partner-CN8-month FE	Yes	Yes	Yes	Yes	Yes
Partner-time FE	Yes	Yes	Yes	Yes	Yes
Product-time FE		HS2	HS4	HS6	CN8

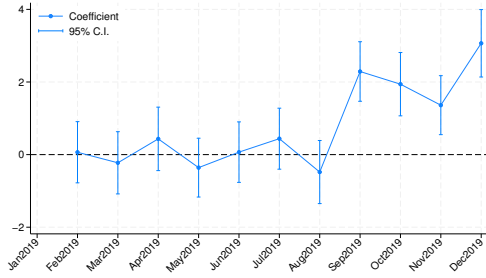
Standard errors clustered at partner-CN8 level in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table reports estimates of the Handley Limao model using the sample of PTA countries of Graziano et al., 2020: Iceland, Norway, Switzerland, Turkey, Mexico, Israel, Chile and South Korea. The dependent variable is the log of UK imports excluding processing trade. The results are based on model where the tail risk $\omega = 1 - (\tau^{PRF} / \tau^{MFN})^4$ is interacted with the log of the Brexit Uncertainty Index. To deal with zeros, I take the hyperbolic sine transformation $\ln[x + (1 + x^2)^{0.5}]$. The model include the interaction of the MFN risk, the current and up to two lags of the log of the BUI. The table reports the sum of the coefficients on the current and all lagged values interactions of BUI and the MFN risk. For Continuity Agreements, the MFN risk and BUI are interacted with a dummy for CA signature. Current and lagged preferential tariffs are included. Columns differ in terms of product-time fixed effects. Column 1 does not include product-time FE. For the other columns the levels are: column 2 at the HS 2-digit, column 3 at the HS 4-digit, column 4 at the HS 6-digit and column 5 at CN 8-digit level.

Table 17: GHL sample, PRF model and Brexit Uncertainty Index

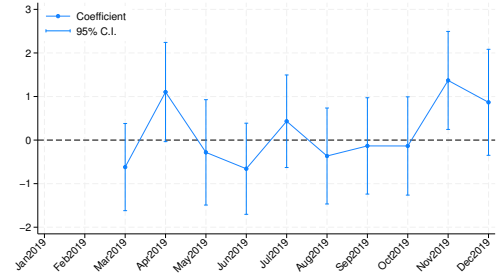
	(1)	(2)	(3)	(4)	(5)
Uncertainty $\times \ln BUI_t$	-1.556*** (0.211)	-1.514*** (0.299)	-2.071*** (0.391)	-2.244*** (0.504)	-3.239*** (0.633)
CA $\times$ Uncertainty $\times \ln BUI_t$	0.919*** (0.225)	0.881*** (0.260)	1.214*** (0.304)	1.126*** (0.357)	1.405*** (0.405)
Tariffs, lags 0-3	Yes	Yes	Yes	Yes	Yes
Observations	514,947	514,488	488,098	415,420	357,312
Partner-CN8-month FE	Yes	Yes	Yes	Yes	Yes
Partner-time FE	Yes	Yes	Yes	Yes	Yes
Product-time FE		HS2	HS4	HS6	CN8

Standard errors clustered at partner-CN8 level in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table reports estimates of the PRF model using the sample of PTA countries of Graziano et al., 2020: Iceland, Norway, Switzerland, Turkey, Mexico, Israel, Chile and South Korea. The dependent variable is the log of UK imports excluding processing trade. The results are based on model where the tail risk $\omega = 1 - (\tau^{PRF}/\tau^{MFN})^4$ is interacted with the share of PRF imports and the log of the Brexit Uncertainty Index. To deal with zeros in the BUI, I take the hyperbolic sine transformation $\ln[x + (1 + x^2)^{0.5}]$. The model include the interaction of the MFN risk, the current and up to two lags of the log of the BUI. The table reports the sum of the coefficients on the current and all lagged values interactions of BUI and the MFN risk. For Continuity Agreements, the MFN risk and the BUI are interacted with a dummy for CA signature. Current and lagged average applied tariffs are included. Columns differ in terms of product-time fixed effects. Column 1 does not include product-time FE. For the other columns the levels are: column 2 at the HS 2-digit, column 3 at the HS 4-digit, column 4 at the HS 6-digit and column 5 at CN 8-digit level.

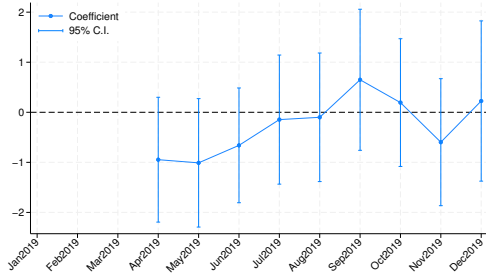
Figure 8: Cohort-time treatment effects of Continuity Agreements



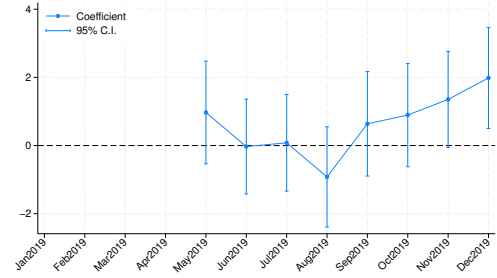
(a) Cohort February 2019



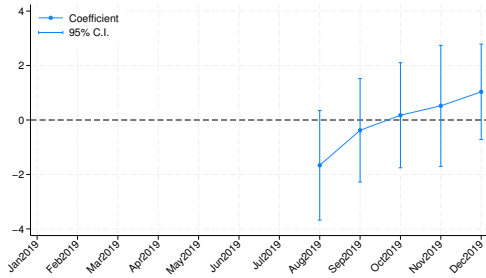
(b) Cohort March 2019



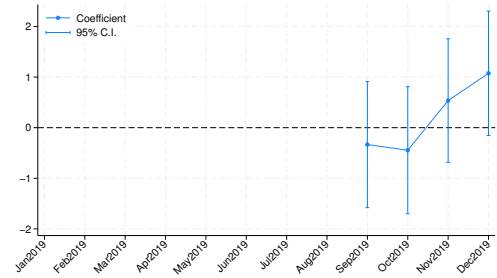
(c) Cohort April 2019



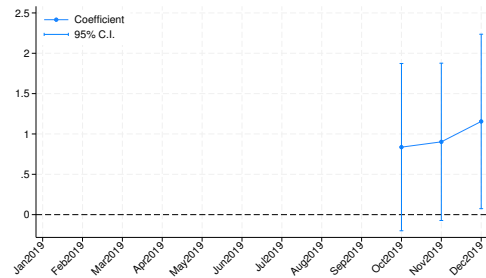
(d) Cohort May 2019



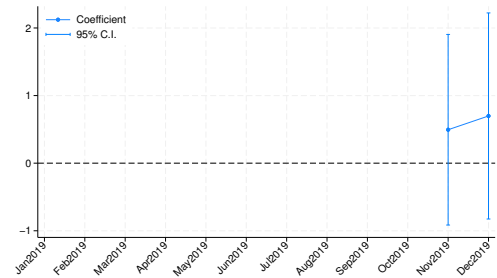
(e) Cohort August 2019



(f) Cohort September 2019



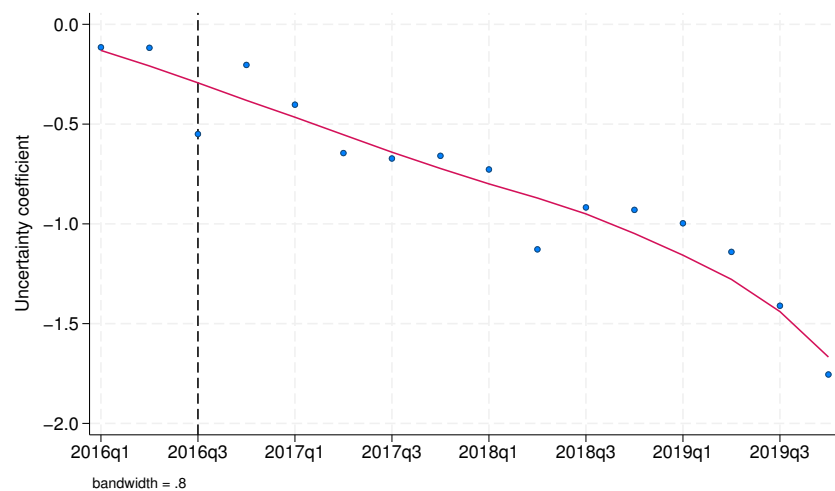
(g) Cohort October 2019



(h) Cohort November 2019

The figure reports the 2019 uncertainty coefficients, the CA effects by cohort-time, together with their 95% confidence interval. Results from the full sample baseline estimation.

Figure 9: Lowess smoother on quarterly uncertainty coefficients



The figure plots LOWESS smoothing results on quarterly average uncertainty coefficients.